



Faculty of Mathematics and Computer Science

Knowledge Discovery (KD)

Steel Industry Knowledge Discovery on Continuous Casting Machine Dataset

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Abstract

This research addresses **predictive maintenance and process optimization in steel manufacturing**, focusing on the continuous casting stage. A **Continuous Casting Machine (CCM)** transforms molten steel into semi finished billets. A key component is the water cooled copper mold **sleeve**, which initiates solidification. Due to high thermal stress and friction, the sleeve degrades over time, making its monitoring important for reducing downtime and maintenance costs.

The main objective is to analyse the **Remaining Useful Life (RUL)** of mold sleeves using industrial data. A preprocessing pipeline was applied to correct this issue by unifying time and date into a single timeline and ordering records by sleeve identifier. A new **RUL** definition was introduced based on total processed steel, ensuring a monotonic and physically consistent measure.

For analysis, numerical variables were discretized into categorical levels (*Low, Medium, High*). The study combines machine learning and knowledge discovery methods. Supervised models, including **Random Forest Regressor (RFR)** and **Random Forest Classifier (RFC)**, were used for baseline prediction and feature importance. Unsupervised methods, including **DBSCAN**, **PCA**, and **Isolation Forest (IF)**, were applied for structure discovery and anomaly detection.

The main knowledge discovery method is **Formal Concept Analysis (FCA)**, which organizes data into concept lattices representing relationships between process variables and equipment states. Association rule mining (Apriori algorithm) was also applied to extract logical dependencies. The analysis was supported by scaling methods implemented using the **ToscanaJ Suite**.

Advanced extensions include **Triadic Formal Concept Analysis (3FCA)**, which introduces operating conditions as a third dimension, enabling analysis of context-dependent relationships. **Temporal Formal Concept Analysis (TCA)** incorporates time to model state transitions and reconstruct lifecycle trajectories.

Overall, combining machine learning with formal concept analysis enables structured interpretation of industrial data and improves understanding of mold sleeve degradation.

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Keywords: (KD) Knowledge Discovery; (KDD) Knowledge Discovery in Databases; (AI) Artificial Intelligence; (ML) Machine Learning; (SL) Supervised Learning; (UL) Unsupervised Learning; (DA) Data Analysis; (CCM) Continuous Casting Machine; (SI) Steel Industry; (SE) Software Engineer; (DM) Data Mining; (FCA) Formal Concept Analysis; (TCA) Temporal Concept Analysis; (RUL) Remaining Useful Life; (RFR) Random Forest Regressor; (RFC) Random Forest Classifier; (DBSCAN) Density-Based Spatial Clustering of Applications with Noise; (PCA) Principal Component Analysis; (IF) Isolation Forest; (AE) Attribute Exploration; (3FCA) Triadic Formal Concept Analysis; (CTS) Conceptual Time System

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1. Introduction

This research paper is driven by my interest in the **Steel Industry (SI)**, more specifically in the **continuous casting process**, which represents one of the most critical stages in modern steel production.

My motivation for selecting this topic is strongly connected to my **professional career**, as I currently work as a **Software Engineer (SE)** in the **industrial automation sector**, specializing in **Industrial Furnaces**. These furnaces operate in a downstream stage of the steel production chain, processing billets and semi-finished products that originate from the **Continuous Casting Machine (CCM)**.

Although my daily work focuses on thermal treatment, reheating and process automation within industrial furnace systems, **understanding the complete lifecycle of steel production is essential for developing a broader and more integrated perspective of the manufacturing process**. The quality of the material entering industrial furnaces is directly influenced by the parameters and operating conditions of the continuous casting process. Defects, inaccuracies, thermal stresses and material inconsistencies produced during casting can propagate through subsequent production stages, ultimately affecting furnace operation and overall plant efficiency.

Studying the **CCM** and the factors that influence its performance is particularly relevant to **my professional development**. From an academic perspective, the selected dataset provides an excellent opportunity to apply techniques from **Artificial Intelligence (AI)/Machine Learning (ML)**, **Knowledge Discovery in Databases (KDD)**, **Data Mining (DM)**, **Formal Concept Analysis (FCA)** and **Temporal Concept Analysis (TCA)**. The dataset contains real industrial measurements related to steel chemistry, casting, cooling and equipment wear, making it suitable for extracting hidden patterns and identifying relationships between variables that can support decision making in industrial environments.

2. Literature Review

The analysis performed in this research combines several disciplines previously mentioned from the fields of **artificial intelligence** and **knowledge discovery**. *Artificial Intelligence (AI)* [16] provides the theoretical framework for developing systems capable of performing tasks that typically require human intelligence, while *Machine Learning (ML)* [3] focuses on constructing models that learn patterns from historical data and make predictions on unseen observations.

Knowledge Discovery in Databases (KDD) [7] represents the overall process of extracting useful, valid, and understandable knowledge from large datasets. *Data Mining (DM)* [10] is considered a core step within the **KDD** process and involves the application of statistical and machine learning techniques to identify patterns, correlations and predictive relationships in data.

To obtain interpretable knowledge from industrial process data, this work also employs *Formal Concept Analysis (FCA)* [8], a mathematical framework for discovering conceptual structures and relationships between objects and attributes. **FCA** enables the construction of **concept lattices**.

Furthermore, *Temporal Concept Analysis (TCA)* [20] extends **FCA** by **incorporating temporal information and state transitions**. This approach allows the investigation of system evolution over time.

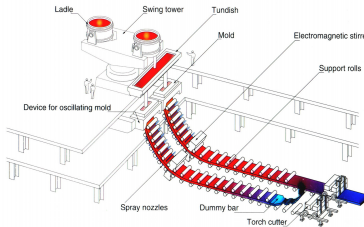


Fig. 1: Image from "Continuous casting practices for steel: Past, present and future" [9]

3. Dataset

3.1. Description and Selection

The dataset used throughout this research was obtained from the **Kaggle** repository *RUL Dataset from Continuous Casting Machine*, published by Yuriy Katser [13].

According to the authors, the dataset was collected from a **real industrial steel production facility** and was specifically prepared for the study of **Remaining Useful Life (RUL)** prediction in continuous casting machine mould sleeves. Before publication, the original production data were processed, cleaned, aggregated and structured to support **ML** and predictive maintenance applications.

Continuous Casting Machine (**CCM**) is an **industrial installation that transforms molten steel into solid billets of a predefined cross-section**, which are subsequently processed through rolling mills to produce finished steel products. One of the most critical components of the **CCM** is the **mould sleeve**, also referred to as the crystallizer sleeve, a water-cooled copper tube responsible for initiating the solidification process. As molten steel comes into contact with the sleeve walls, a primary solid shell is formed, allowing the billet to maintain its geometry while moving through the casting process.

The operational degradation of mould sleeves represents a significant industrial challenge. Defects can propagate through downstream manufacturing stages and negatively affect rolling quality, production efficiency and operational costs. Mould sleeves service life is approximately 17,000 tons for billets of section (180 * 180) mm and 13,000 tons for billets of section (150 * 150) mm. Premature sleeve replacement increases maintenance costs, equipment downtime and operational risks.

A particularly important attribute of the dataset is the **RUL** variable, which value is derived from the *resistance, tonn* attribute by calculating, for each unique combination of sleeve, crystallizer and casting stream, the difference between the maximum recorded resistance (corresponding to sleeve failure) and the current resistance value. Consequently, the **RUL** attribute represents the remaining production capacity of the mould sleeve before replacement becomes necessary.

The wide range of the available attributes makes the dataset particularly appropriate for *Formal Concept Analysis (FCA)*, *Temporal Concept Analysis (TCA)* and the construction of conceptual scales within **ToscanaJ**, for both predictive and explainable knowledge extraction.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y
date	workpiece	steel_type	doc	requi	cast_in	ro	workpiece	alloy_type	steel_weight	slag_weight	metal_resl	steel_v	grab1_nur	steel_temj	grab2_nur	resistance	swing	freq	crystallize	alloy_spee	water_con	water_tem	water_con	water_g
1/5/2020	144.9	Arm240	DOC 3402		4	150x150	open		145.3	1.8	0.4	145	4		5	3316	180	9	3	1950	9	360	400	1
1/5/2020	165.9	St3sp	Contract		10	150x150	open		166.3	1.8	0.4	166	7	1563	5	4955	180	10	3	1950	9	360	420	1
1/5/2020	168	Arm240	DOC 3402		5	150x150	open		168.4	1.8	0.4	168	7	1579	5	3345	180	10	3	1950	9	340	400	1
1/5/2020	170.1	St3sp	Contract		7	150x150	open		170.5	1.8	0.4	170	12	1563	5	3400	180	10	3	1950	9	360	420	1
1/5/2020	163.8	St3sp	Contract		12	150x150	open		164.2	1.8	0.4	164	12	1565	5	3536	180	10	3	1950	9	360	420	1
1/5/2020	163.8	St3sp	Contract		14	150x150	open		164.2	1.8	0.4	164	4	1568	5	3591	180	10	3	1950	9	360	420	1
1/5/2020	163.8	Arm240	DOC 3402		5	150x150	open		164.2	1.8	0.4	164	7	1573	3	3123	165	10	3	1950	9	320	400	1
1/5/2020	107.428	Arm240	DOC 3402		2	150x150	close		135.828	1.8	0.4	135		1579	4	3209	180	7	2	1950	9	200	250	1
1/5/2020	161.7	St3sp	Contract		6	150x150	open		162.1	1.8	0.4	162	3	1568	5	3373	180	10	3	1950	9	360	410	1
1/5/2020	165.9	St3sp	Contract		13	150x150	open		166.3	1.8	0.4	166	5	1566	5	3564	180	10	3	1950	9	360	420	1
1/5/2020	174.3	St3sp	Contract		15	150x150	open		174.7	1.8	0.4	174	7	1566	5	3620	180	10	3	1950	9	360	420	1
1/5/2020	165.9	St3sp	Contract		8	150x150	open		166.3	1.8	0.4	166	5	1564	5	3427	180	10	3	1950	9	360	420	1
1/5/2020	165.9	St3sp	Contract		9	150x150	open		166.3	1.8	0.4	166	5	1565	5	3454	180	10	3	1950	9	360	420	1
1/5/2020	165.9	St3sp	Contract		10	150x150	open		166.3	1.8	0.4	166	7	1563	5	3482	180	10	3	1950	9	360	420	1
1/5/2020	168	St3sp	Contract		11	150x150	open		168.4	1.8	0.4	168	3	1564	5	3509	180	10	3	1950	9	360	420	1
1/5/2020	157.449	Arm240	DOC 3402		1	150x150	close		158.449	1.8	0.4	158	12	1584	4	3184	165	10	3	1950	8	210	250	1
1/5/2020	159.6	Arm240	DOC 3402		1	150x150	open		160.6	1.8	0.4	160	3	1583	5	3236	180	9	2	1950	9	260	320	1
1/5/2020	170.1	Arm240	DOC 3402		2	150x150	open		170.5	1.8	0.4	170	12	1582	5	3265	180	9	2	1950	9	320	350	1
1/5/2020	163.8	Arm240	DOC 3402		3	150x150	open		164.2	1.8	0.4	164	5	1572	5	3293	180	9	3	1950	9	350	400	1
1/5/2020	168	St3sp	Contract		16	150x150	open		168.4	1.8	0.4	168	3	1565	5	3646	180	10	3	1950	9	360	420	1
1/5/2020	170.1	St3sp	Contract		17	150x150	open		170.5	1.8	0.4	170	12	1566	5	3673	180	10	3	1950	9	360	420	1
1/5/2020	163.8	St3sp	Contract		18	150x150	open		164.2	1.8	0.4	164	5	1565	5	3700	180	10	3	1950	9	360	420	1
1/5/2020	157.449	Arm240	DOC 3402		1	150x150	close		158.449	1.8	0.4	158	12	1584	4		165	10	3	1950	8	210	250	1
1/5/2020	165.9	St3sp	Contract		9	150x150	open		166.3	1.8	0.4	166	5	1565	5	4928	180	10	3	1950	9	360	420	1

Fig. 2: Dataset Excel Visualization

3.2. Dataset Attributes

Table 1 summarizes the main categories of attributes contained in the dataset.

Table 1: Main dataset attributes and their descriptions

Attribute(s)	Description
date	Date of the casting operation
time_temperature_measurement1, time_temperature_measurement2, sample_time_continuous_caster	Time-related process measurements
workpiece_weight, steel_weight_theoretical, steel_weight	Actual and theoretical steel mass processed during a cast
steel_type	Steel grade produced during the casting process
doc_requirement	Production specification or quality requirement associated with the cast
cast_in_row	Sequential number of the cast within a production campaign
workpiece_slice_geometry	Cross-sectional geometry of the produced billet (e.g., 150×150 mm)
alloy_type	Alloy production mode or casting configuration
slag_weight_close_grab1, metal_residue_grab1, residuals_grab2, technical_trim	Process by-products and material losses generated during casting
grab1_num, grab2_num	Sampling or process operation identifiers
steel_temperature_grab1, temperature_measurement1, temperature_measurement2	Steel temperature measurements recorded at different process stages
resistance	Accumulated tonnage processed by the mould sleeve, representing equipment wear
swing_frequency	Oscillation frequency of the crystallizer during casting
crystallizer_movement	Oscillation amplitude or movement of the crystallizer
alloy_speed	Casting speed of the molten steel
water_consumption, water_temperature_delta	Primary cooling system operating parameters
water_consumption_secondary cooling_zone_1–3	Cooling water flow rates in the secondary cooling zones
kind, quantity	Production batch characteristics and associated quantities
Ce, C, Si, Mn, S, P, Cr, Ni, Cu, As, Mo, Nb, Sn, Ti, V, Al, Ca, N, Pb, Mg, Zn	Chemical composition of the steel expressed as elemental percentages
sleeve	Unique identifier of the mould sleeve
num_crystallizer	Identifier of the crystallizer used during casting
num_stream	Casting strand identifier
RUL	RUL of the mould sleeve, expressed in tons of steel production remaining before replacement. This attribute represents the target variable used throughout the analysis

3.3. Dataset Analysis

The original **RUL** attribute provided in the dataset **was found to be difficult to interpret** as a true remaining useful life measure. During the exploratory analysis, several sleeves exhibited multiple **RUL** countdown sequences, where the **RUL** value reached zero and later increased again. This behaviour is inconsistent with the expected lifecycle of a physical component, since a sleeve cannot regain useful life after it has been consumed. According to the dataset documentation, the original **RUL** was derived from resistance measurements rather than from the actual cumulative usage of a sleeve. Therefore, a new **RUL** metric was **calculated based on the total amount of steel processed by each sleeve over time**. This produced a monotonically decreasing value that better represents the progressive wear of the sleeve throughout its operational lifecycle.

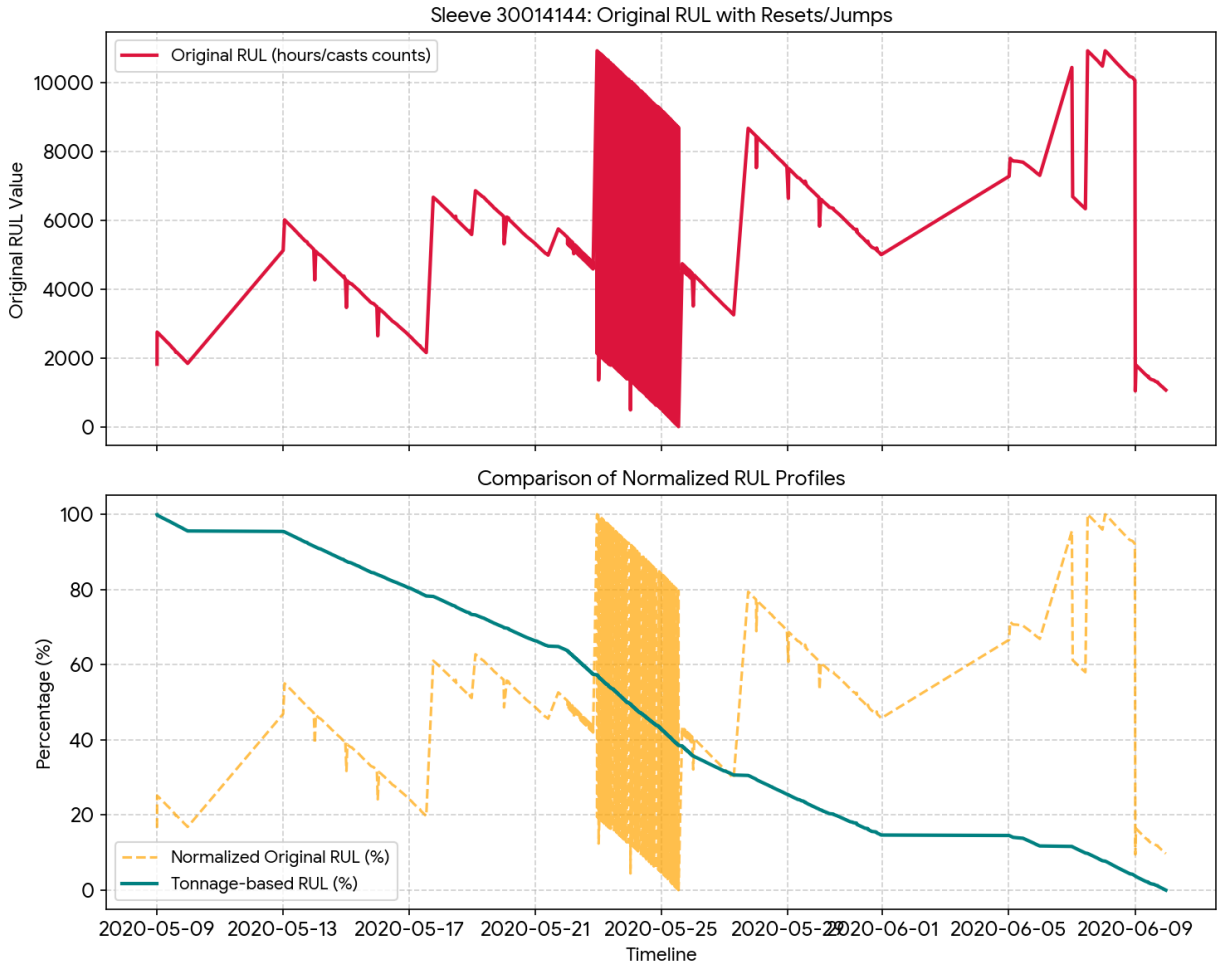


Fig. 3: Original **RUL** unclarity over a normal **RUL**

3.4. Dataset Preprocessing

The preprocessing stage was designed to **reconstruct the operational history of each sleeve** and prepare the data for lifecycle analysis and Formal Concept Analysis (FCA). First, the separate date and time fields were combined into a single timestamp to create a complete chronological record of the production process. The dataset was then sorted by sleeve identifier and timestamp so that all events belonging to the same sleeve could be analysed in their correct temporal order.

For each sleeve, a cumulative tonnage value was calculated by continuously summing the amount of processed steel. The maximum cumulative tonnage reached by a sleeve was considered its total operational capacity, and a new Remaining Useful Life (RUL) value was computed as the difference between this maximum capacity and the current cumulative tonnage.

To allow comparison between sleeves of different lifetimes, the calculated RUL was normalized into a **percentage scale** ranging from 100% at the beginning of the lifecycle to 0% at the end. Based on this, each observation was assigned to one of four health categories: Critical (0–25%), Low (25–50%), Medium (50–75%), and Healthy (75–100%).

Only the most relevant process, production and equipment attributes were retained for further analysis. To facilitate FCA and data mining, continuous numerical attributes were discretized into three categories (Low, Medium, High) and encoded into numerical representations. This transformation produced a dataset suitable for concept lattice construction, scale generation and lifecycle pattern discovery.

Table 2: Additional attributes generated during dataset preprocessing

Attribute	Description
datetime_combined	Combined date and time of continuous casting sample acquisition
calculated_RUL_tons	Estimated remaining useful life of the caster expressed in tons of steel processed
RUL_percentage	Remaining useful life expressed as a percentage of nominal lifetime
*_BIN	Discretized representation of continuous numerical variables (Low, Medium, High) used for rule extraction and interpretability
*_ENCODED	Integer-encoded representation of categorical and discretized variables for machine learning pipelines
steel_type, alloy_type, work-piece_slice_geometry	Categorical process descriptors of steel grade, alloy classification and slab geometry
sleeve, num_crystallizer, num_stream	Equipment configuration identifiers for the casting setup
temperature_measurements	Multiple temperature sensor readings used for process monitoring and validation
cooling_related variables	Includes water consumption, water temperature delta and derived cooling behavior descriptors
process_kinematics	Includes swing frequency, crystallizer movement and alloy speed describing caster dynamics
steel_weight, steel_temperature_grab1, resistance	Primary process measurements recorded during casting operations

	date,time_combined	steel_type	cast_in_c	r_workpiece	alloy_type	steel_weight	steel_temp	resistance	swing_fre	crystallize_alloy	spk_water	cor_water	temp	temp_op	temp_p	sleeve	num_cyl	num_strc	calculated_RUL	perc_RUL	RUL_FAI	BCIN	V_FAI	V_CAI	r_cast_in	r_cast_in	steel_weight	steel_temp	AA
2	3/5/2020 0:15 Arm000	1180L80	open			161.172	1587	1789	180	7	2	2155	8	1544	1549	00011717	1	15688.7	98.9939	Healthy				3	Low	0	Medium	1	High
3	3/5/2020 1:16 Arm000	1180L80	open			169.344	1583	1819	180	7	2	2155	8	1548	1548	00011717	1	15588.6	97.8367	Healthy				3	Low	0	High	2	High
4	3/5/2020 2:04 Arm000	1180L80	open			166.324	1584	1818	180	7	2	2155	8	1546	1543	00011717	1	15572.44	96.8981	Healthy				3	Low	0	High	2	High
5	3/5/2020 2:54 Arm000	4180L80	open			169.344	1581	1876	180	7	2	2155	9	1544	1545	00011717	1	15553.09	95.84138	Healthy				3	Low	0	High	2	High
6	3/5/2020 3:53 Arm000	5180L80	open			163.296	1580	1954	180	7	2	2155	9	1545	1545	00011717	1	15189.8	94.82201	Healthy				3	Low	0	High	2	High
7	3/5/2020 4:25 Arm000	6180L80	open			163.296	1578	1931	180	7	2	2155	9	1548	1548	00011717	1	15026.6	93.80264	Healthy				3	Low	0	High	2	High
8	3/5/2020 5:08 Arm000	7180L80	open			169.344	1609	1958	180	7	2	2155	9	1548	1546	00011717	1	14857.16	92.74551	Healthy				3	Low	0	High	2	High
9	3/5/2020 6:04 Arm000	8180L80	open			169.344	1676	1975	180	7	2	2155	9	1549	1549	00011717	1	14697.61	91.6889	Healthy				3	Low	0	High	2	High
10	3/5/2020 6:38 Arm000	9180L80	open			166.32	1570	2015	180	7	2	2155	9	1545	1545	00011717	1	14521.49	90.65014	Healthy				3	Low	0	High	2	High
11	3/5/2020 7:24 Arm000	10180L80	open			169.344	1575	2046	180	7	2	2155	9	1545	1546	00011717	1	14382.15	89.59301	Healthy				3	Low	0	High	2	High
12	3/5/2020 8:14 Arm000	11180L80	open			169.344	1574	2073	180	7	2	2155	9	1549	1548	00011717	1	14182.81	88.53588	Healthy				3	Low	0	High	2	High
13	3/5/2020 9:00 Arm000	12180L80	open			163.296	1575	2100	180	7	2	2155	9	1553	1550	00011717	1	14019.51	87.51651	Healthy				3	Low	0	High	2	High
14	3/5/2020 9:43 Arm000	13180L80	open			166.32	1573	2127	180	7	2	2155	9	1549	1545	00011717	1	13853.19	86.47656	Healthy				3	Low	0	High	2	High
15	3/5/2020 10:21 Arm000	14180L80	open			166.32	1571	2151	180	7	2	2155	9	1546	1546	00011717	1	13616.87	85.44001	Healthy				3	Low	0	High	2	High
16	3/5/2020 11:02 Arm000	15180L80	open			163.296	1571	2179	180	7	2	2155	9	1544	1544	00011717	1	13525.37	84.42064	Healthy				3	Low	0	High	2	High
17	3/5/2020 11:53 Arm000	16180L80	open			163.296	1573	2206	180	7	2	2155	9	1545	1547	00011717	1	13360.26	83.40127	Healthy				3	Low	0	High	2	High
18	3/5/2020 12:34 Arm000	17180L80	open			169.344	1569	2233	180	7	2	2155	9	1547	1545	00011717	1	13190.93	82.34414	Healthy				3	Low	0	High	2	High
19	3/5/2020 13:18 Arm000	18180L80	open			166.32	1573	2260	180	7	2	2155	9	1549	1551	00011717	1	13024.61	81.30589	Healthy				3	Low	0	High	2	High
20	3/5/2020 14:05 Arm000	19180L80	open			163.296	1571	2285	180	7	2	2155	9	1548	1550	00011717	1	12861.32	80.26652	Healthy				3	Low	0	High	2	High
21	3/5/2020 14:49 Arm000	20180L80	open			166.32	1569	2315	180	7	2	2155	9	1547	1545	00011717	1	12699	79.24827	Healthy				3	Low	0	High	2	High
22	3/5/2020 15:30 Arm000	21180L80	open			169.344	1572	2342	180	7	2	2155	9	1549	1548	00011717	1	12525.65	78.19115	Healthy				3	Medium	1	High	2	High
23	3/5/2020 16:21 Arm000	22180L80	open			163.296	1578	2369	180	7	2	2155	9	1545	1548	00011717	1	12362.36	77.17177	Healthy				3	Medium	1	High	2	High

Fig. 4: Preprocessed Dataset Excel Visualization

4. First Information Layer

In this part of the project, a complete **Data Analysis (DA)** and **Knowledge Discovery (KD)** process on the preprocessed **CCM** dataset was performed. The main goal was to understand the data, explore important patterns and build machine learning models that can help explain the Remaining Useful Life (**RUL**) of the sleeve.

The dataset used in this analysis is the preprocessed one previously exposed in section 3. This means that the numerical sensor values were already normalized using **StandardScaler** and the **FCA** necessary discretization bins were already created. The dataset also contained the target variables '**RUL**' and '**RUL_class**', which were used for regression and classification tasks.

4.1. Data Plots

The first analysis was performed by creating a descriptive report. The following pages will expose the size of the dataset, **RUL** values and class distribution and other relevant data.

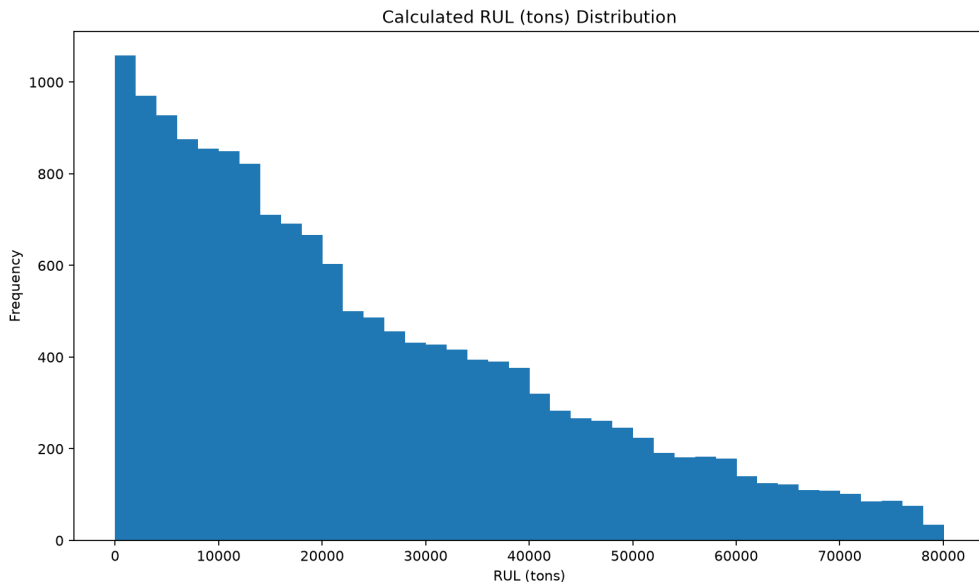


Fig. 5: **RUL** values distribution

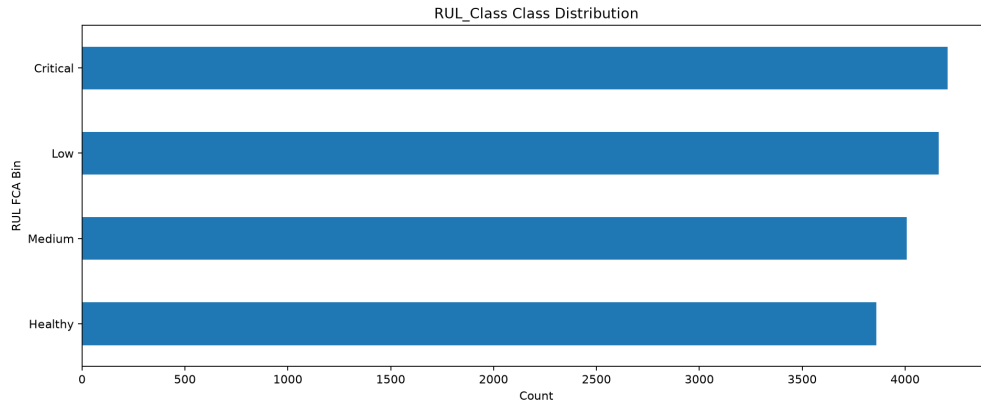


Fig. 6: RUL classes distribution

Also, the average RUL was analyzed based on different industrial groups, such as workpiece geometry, stream, crystallizer and alloy type. This helped in understanding how the RUL value changes across different production conditions.

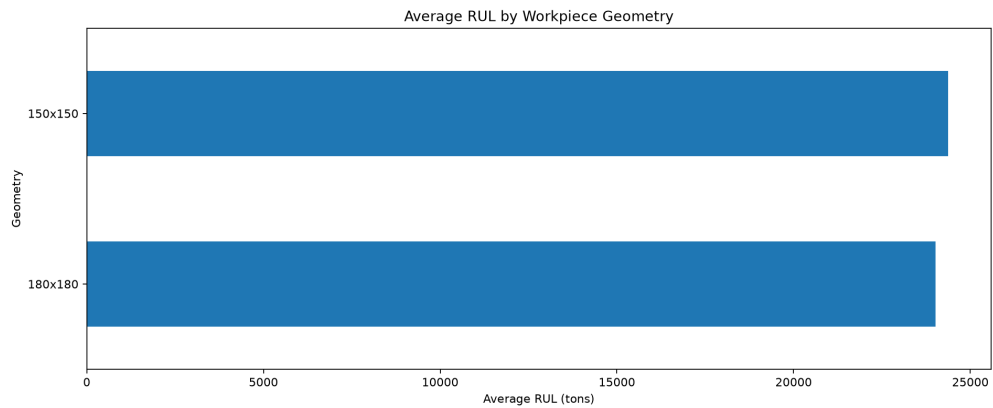


Fig. 7: Average RUL per geometry distribution

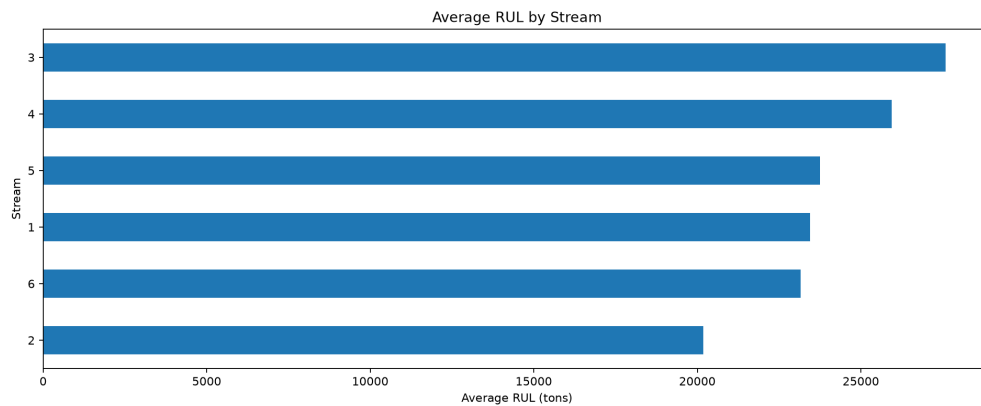


Fig. 8: Average RUL per stream distribution (equipment identifiers)

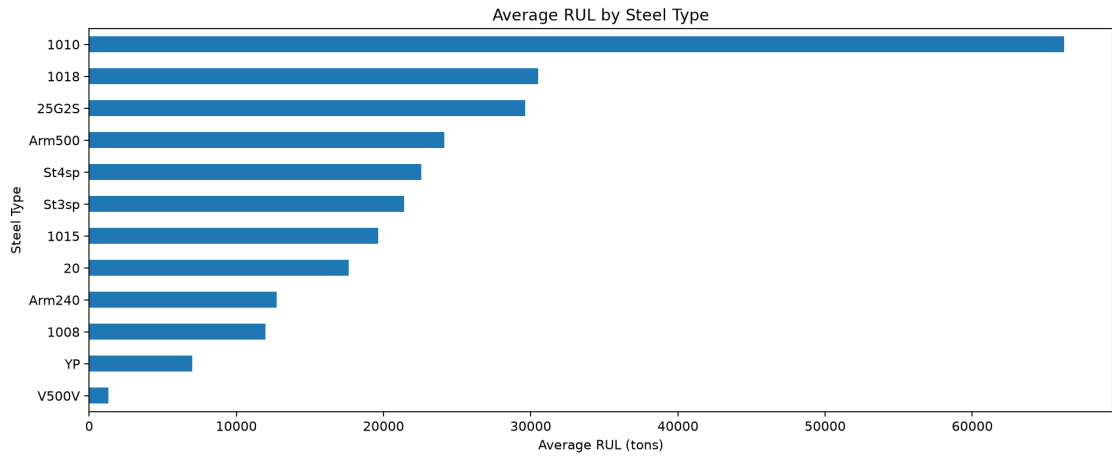


Fig. 9: Average RUL per steel type

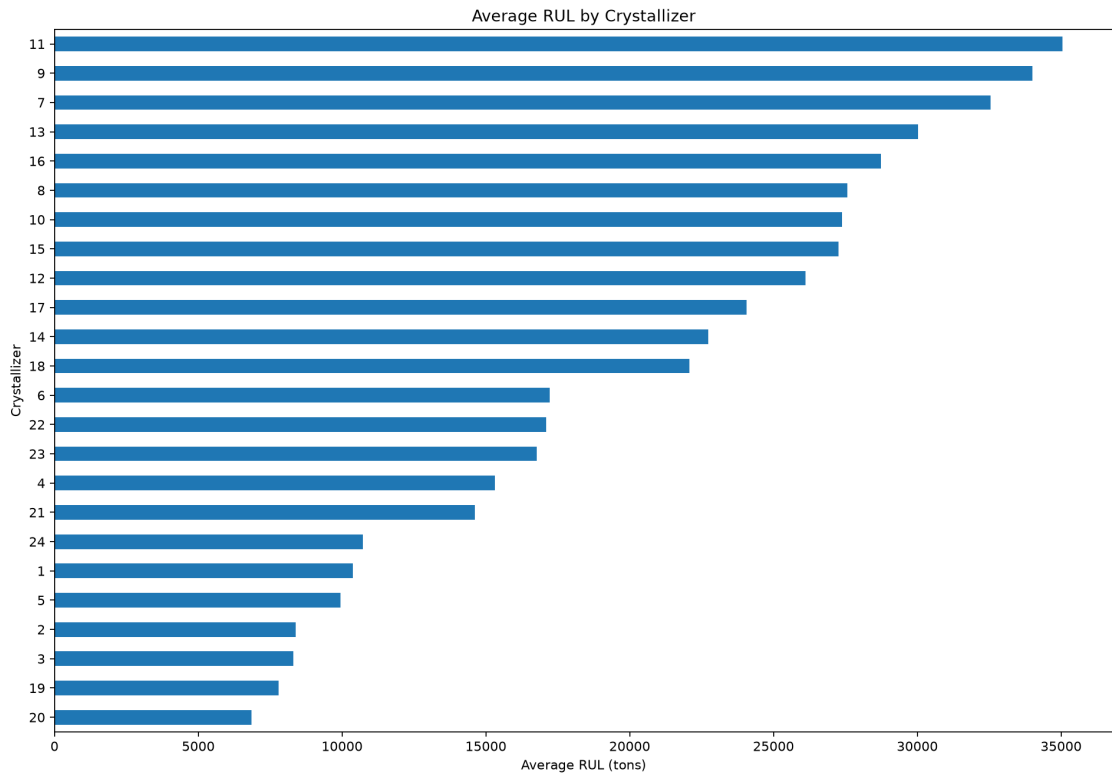


Fig. 10: Average RUL per crystallizer distribution (equipment identifiers)

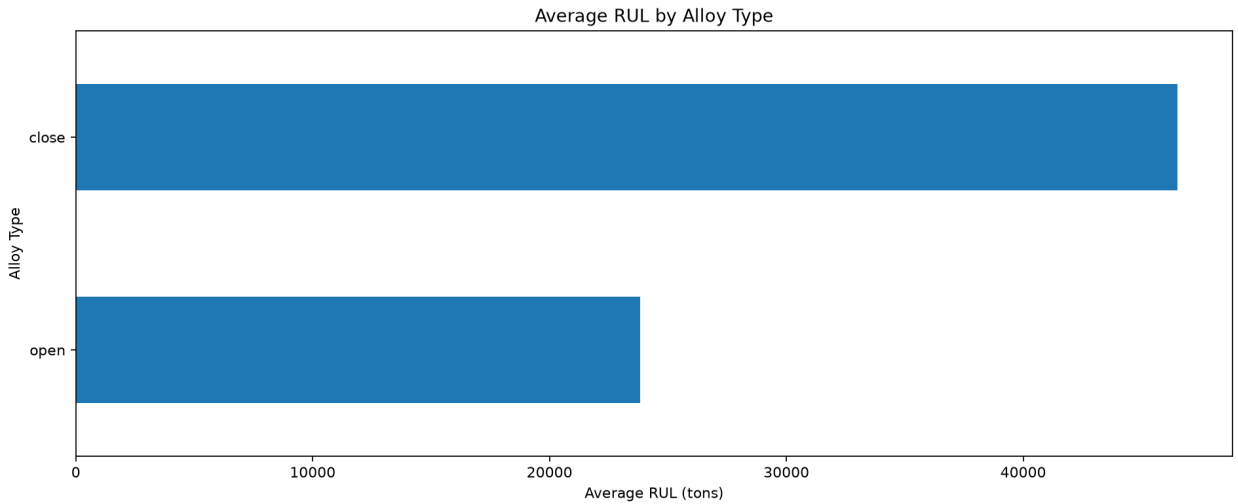


Fig. 11: Average RUL per alloy type

A histogram of the RUL distribution was generated to observe how the target values are spread.

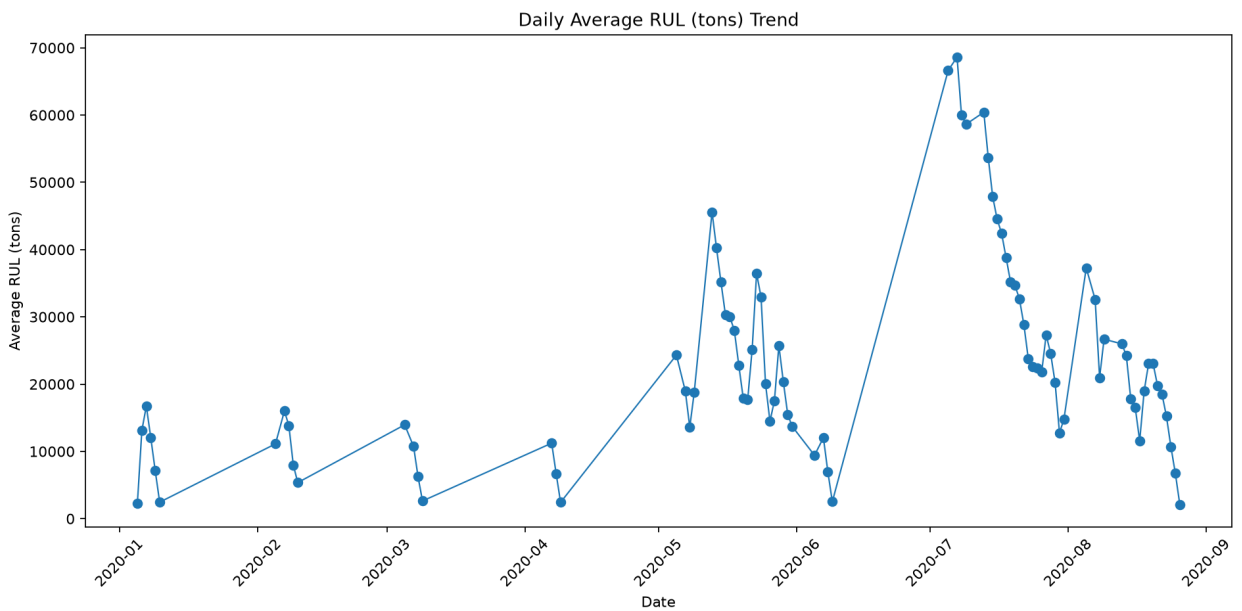
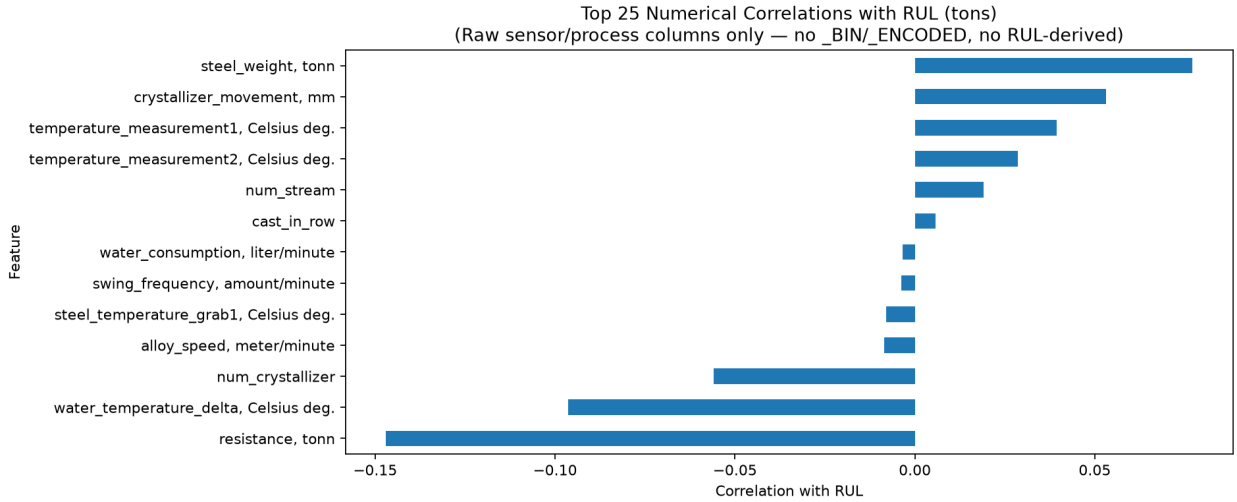


Fig. 12: Average RUL distribution history

Another important part of the exploratory analysis was the calculation of numerical correlations between the available features and the RUL target. This helped identify which variables are most strongly related to RUL. The following plot is used to visually check how sensor values change in relation to sleeve lifetime.

Fig. 13: Top numerical correlations **RUL** feature

4.2. Supervised learning

For supervised learning (**Supervised Learning (SL)**), **RUL** was used as a **regression target**. **SL** is a **machine learning approach** where a model is trained on **labeled data**, meaning each input example is paired with a known output, and the goal is to learn a mapping from inputs to outputs that can be used to predict the output for unseen data [3]. A **Random Forest Regressor (Random Forest Regressor (RFR))** was trained to predict **RUL** using process and sensor features. The **RFR** builds multiple decision trees on random subsets of the training data and features and the final prediction is obtained by averaging the outputs of all trees [4]. The dataset was split into training and testing sets and the model was evaluated using **Mean Absolute Error (MAE)**, **Root Mean Squared Error (rmse)** and **R-squared (R^2)**.

Metric	Value	Definition
R^2	0.90	Measures how well the model explains the variance in the target variable. A value close to 1 indicates a strong fit [11]
MAE	2748	The average absolute difference between predicted and actual values [5]
RMSE	5790	Square root of the average squared differences between predicted and actual values (penalizes large errors more strongly) [5]

Table 3: Evaluation metrics for the **RFR** model

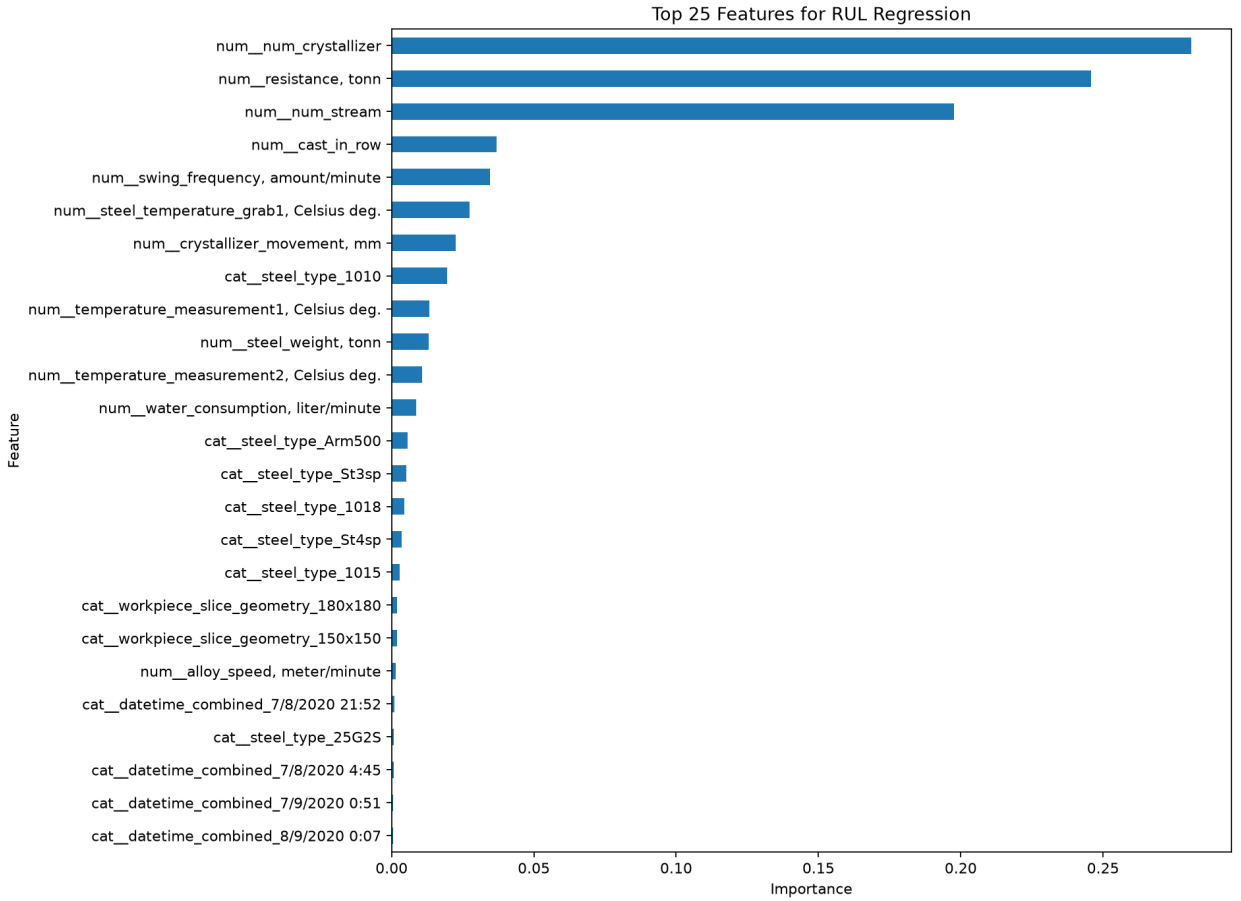


Fig. 14: Feature importance analysis for the RFR model

The previous plot determines which input features had the strongest effect on RUL predictions. It also makes the RFR model easier to interpret.

For classification, 'RUL_class' was used as the target variable. A **Random Forest Classifier** (*Random Forest Classifier (RFC)*) was trained to predict sleeve condition categories based on RUL classes. It constructs multiple decision trees on random subsets of the training data and features and the final class prediction is obtained by majority voting across all trees [4].

Metric	Value	Definition
Accuracy	0.6856	The proportion of correctly classified samples out of all predictions, measures overall correctness [15]
Precision (macro)	0.6865	The proportion of correctly predicted positive samples among all predicted positive samples [15]
Recall (macro)	0.6859	The proportion of correctly predicted positive samples among all actual positive samples [15]
F1-score (macro)	0.6862	The harmonic mean of precision and recall, providing a balanced measure between both metrics, especially useful for imbalanced datasets [15]

Table 4: Evaluation metrics for the RFC model

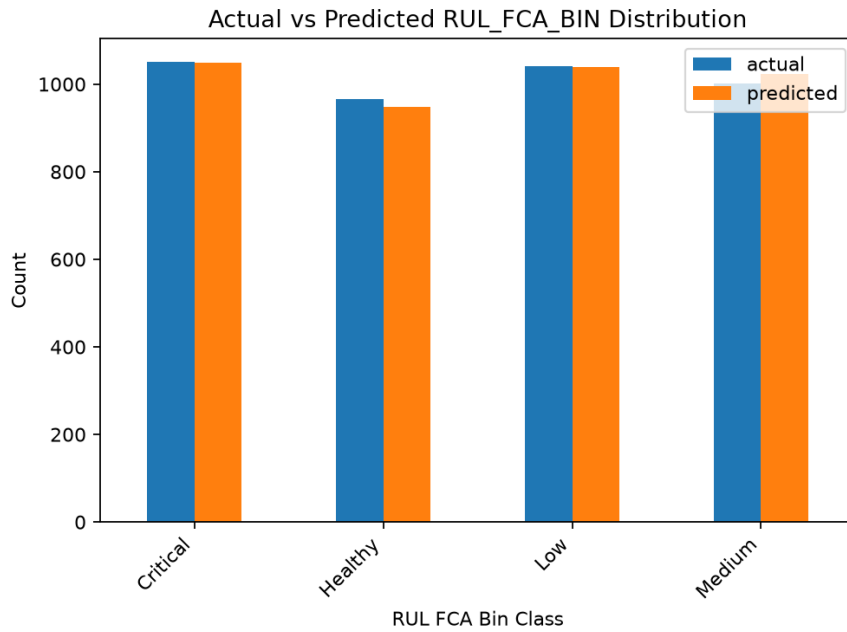


Fig. 15: Classification actual and predicted distributions for the **RFR** model

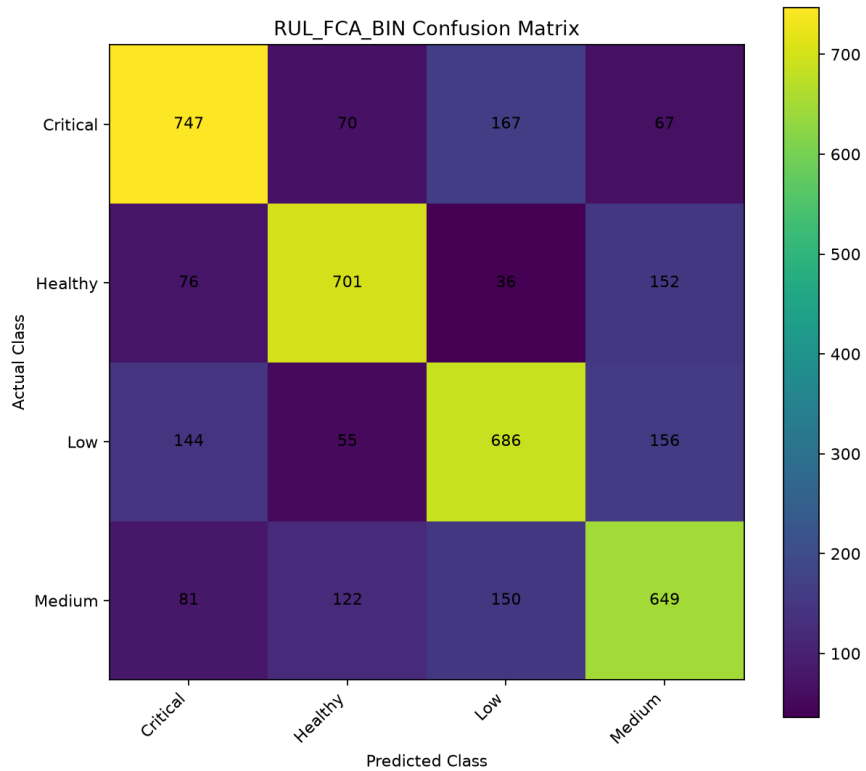


Fig. 16: Confusion matrix for the **RFR** model

4.3. Unsupervised learning

In addition to supervised learning (SL), some unsupervised methods were applied. **Unsupervised learning (Unsupervised Learning (UL))** is a ML approach that works with unlabelled data and Its goal is to discover hidden patterns, structures, relationships or groups within the data without using predefined target values [3]. **K-Means Clustering** was used to group similar production cases. K-Means is an unsupervised clustering algorithm that partitions data into a predefined number of clusters (K), in this case 4 (according the the 4 RUL classes defined) [3]. **Density-Based Spatial Clustering of Applications with Noise (Density-Based Spatial Clustering of Applications with Noise (DBSCAN))** was also applied to detect **dense clusters** and **noise points** [6]. **Principal Component Analysis (Principal Component Analysis (PCA))** was used to reduce the dataset to two main components for visualization and as a base for UL algorithms. PCA is a dimensionality reduction technique that transforms a dataset into a smaller set of uncorrelated variables called **principal components** [12]. These components capture most of the variance in the data, making it easier to visualize and analyse complex datasets while reducing information loss.

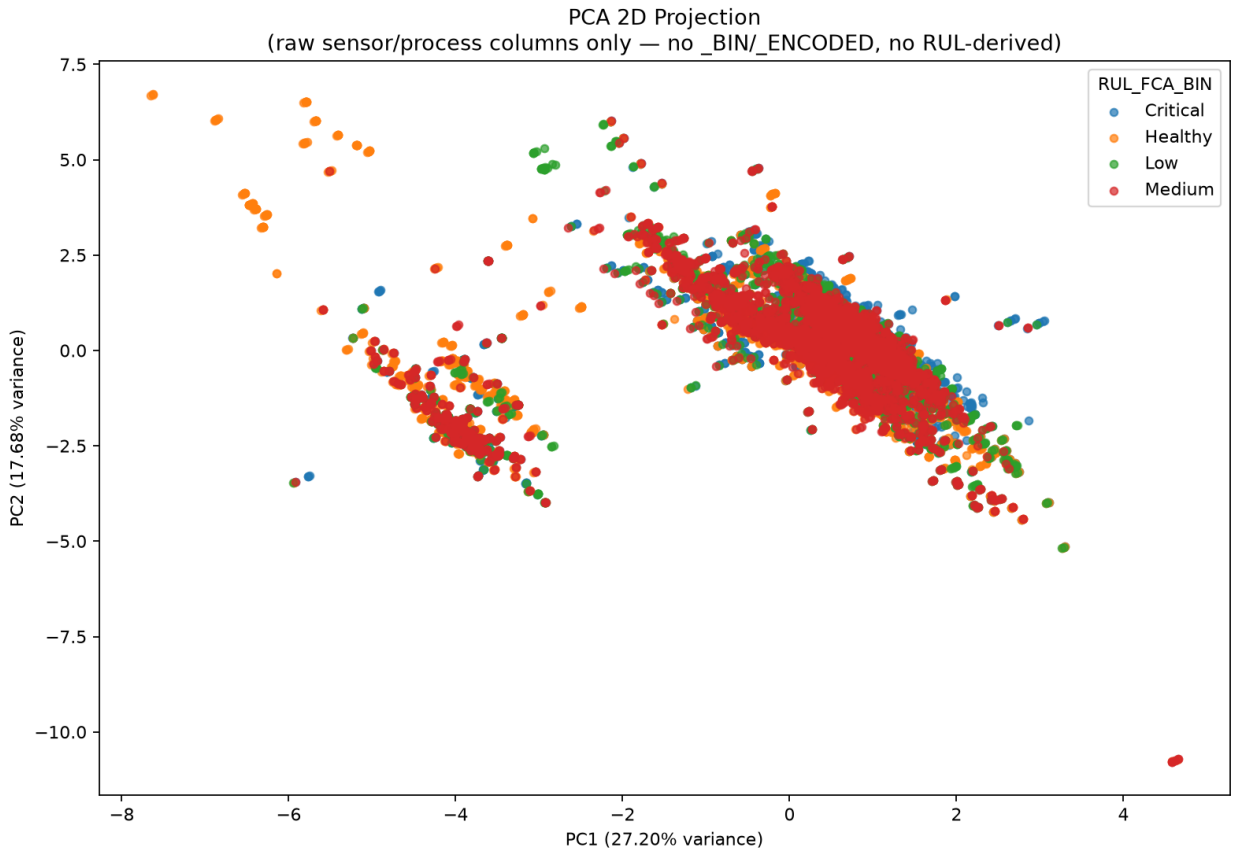


Fig. 17: PCA over RUL classes

Principal Component	Explained Variance Ratio
PC1	0.2720
PC2	0.1768

Table 5: Explained variance ratio of the first two principal components

The first principal component (*PC1*) explains 27.20% of the total variance, while the second principal component (*PC2*) explains 17.68%. Together, the two components account for 44.88% of the total variance. Although this captures a substantial portion of the dataset's variability, more than half of the variance remains outside the two-dimensional projection. Therefore, the **PCA visualization provides only a partial representation of the underlying data structure**. **K-Means** forces the data into a predefined number of clusters, which may not align with the intrinsic structure of the data. **DBSCAN** primarily groups most observations into one large cluster, with a smaller separate cluster, several tiny dense groups, and a small proportion of points classified as noise. Overall, these results suggest that a two-dimensional representation may not be sufficient to fully capture the complexity of the dataset and clustering approaches operating in higher dimensional feature spaces may provide more reliable results.

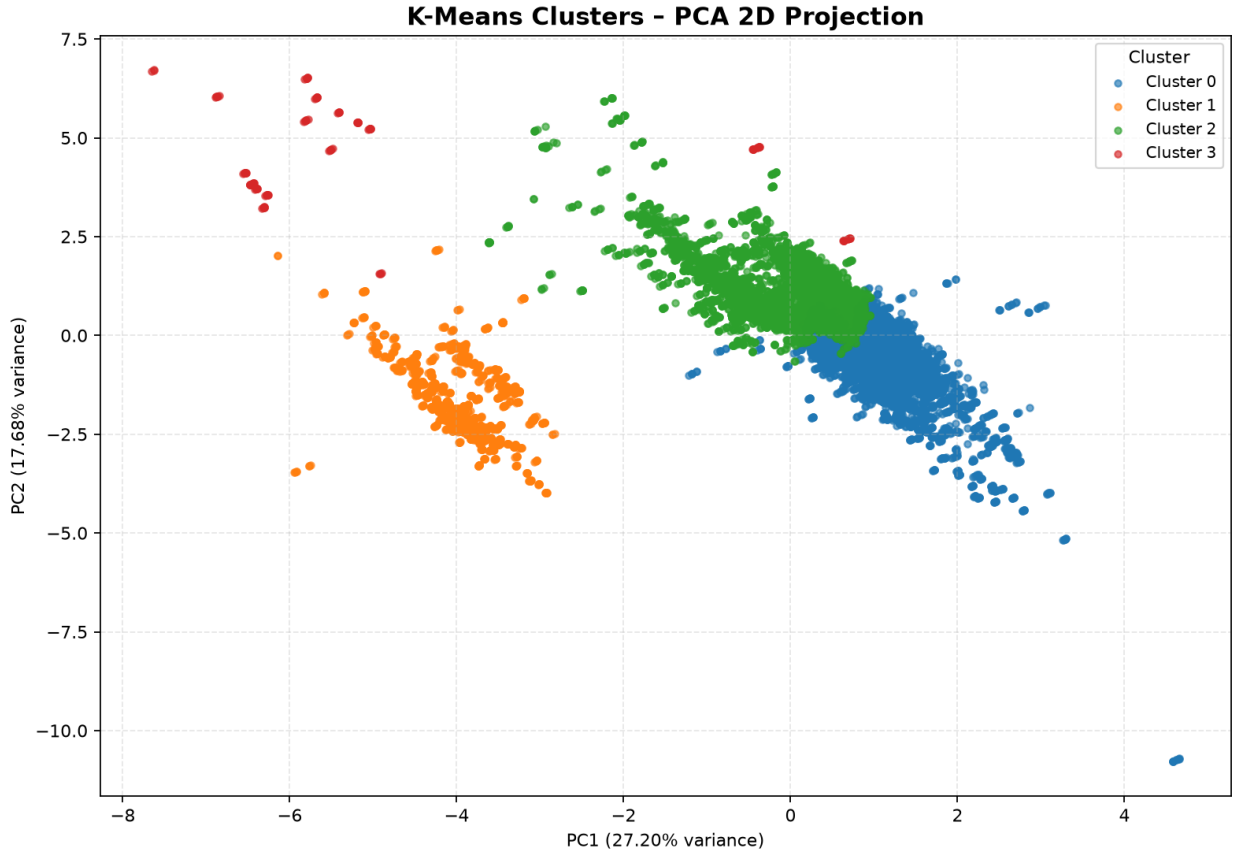


Fig. 18: K-Means Clustering over PCA

4.4. Anomaly Detection

An **Isolation Forest (Isolation Forest (IF))** was used to find unusual production cases. **IF** is an unsupervised anomaly detection algorithm that identifies outliers by isolating data points through random feature splits [14].

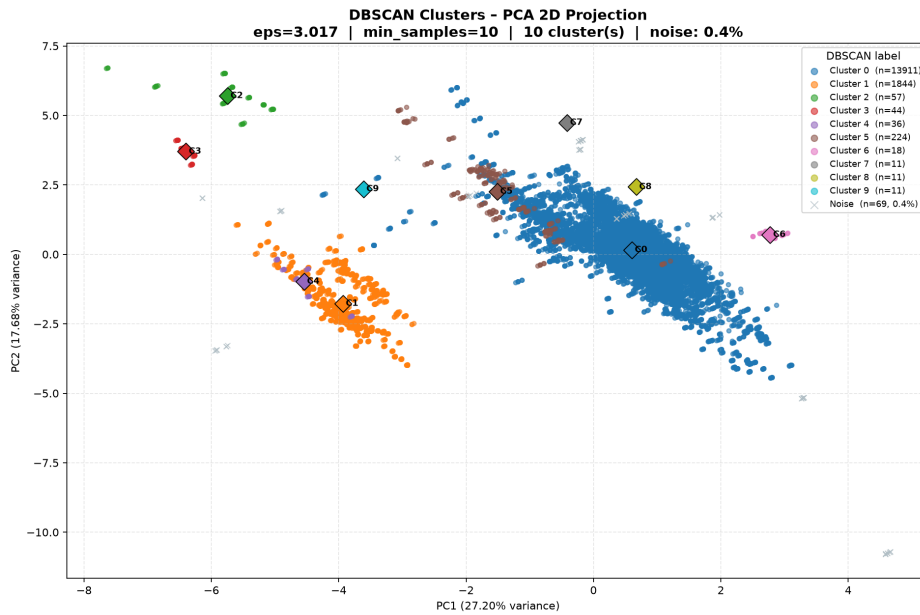


Fig. 19: IF anomaly over scatter PCA

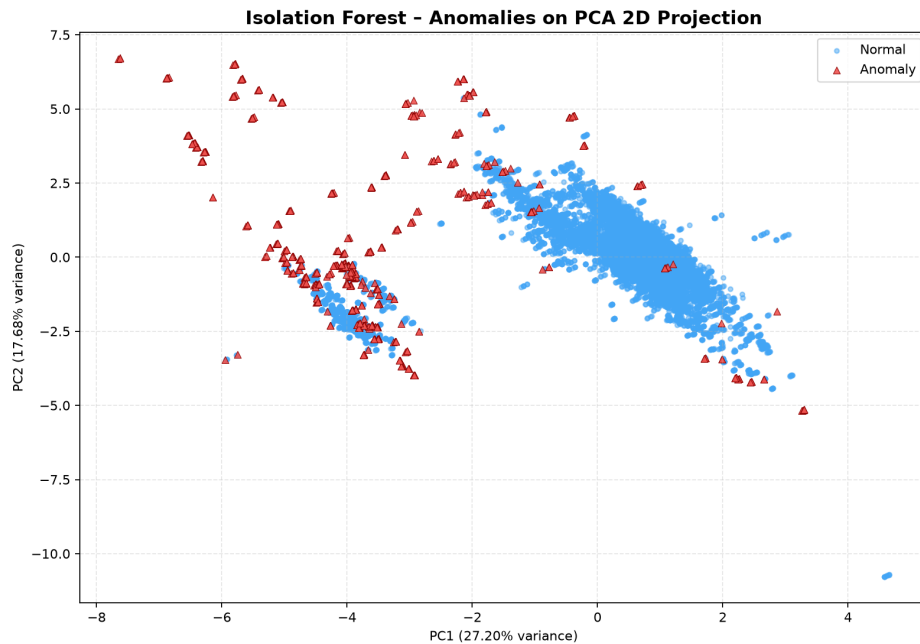


Fig. 20: IF anomaly heat map over PCA

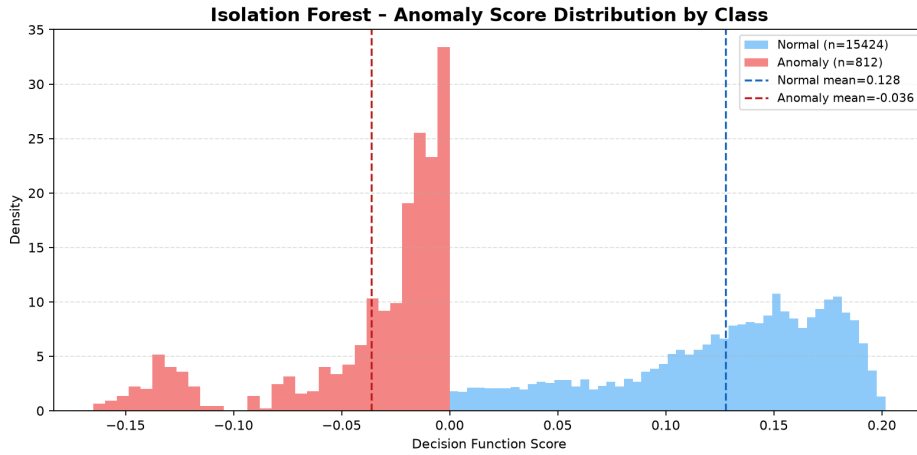


Fig. 21: IF score distribution

These cases may represent **abnormal behaviour** or **faster sleeve degradation**.

4.5. Association Rule Mining

Association rule mining was applied using Formal Concept Analysis [FCA](#) binary columns and the 'RUL_class' column. There, the **Apriori** algorithm was used. It is an unsupervised learning method used in association rule mining [1], used to find frequent item sets in large datasets and then generates rules that describe how items co-occur together. The resulting rules were ranked based on lift and confidence. **Confidence** measures the reliability of an association rule, indicates how often the rule is correct when the condition is satisfied. **Lift** measures the strength of an association rule compared to random chance, a lift value greater than 1 indicates a positive association, while a value close to 1 suggests no meaningful relationship.

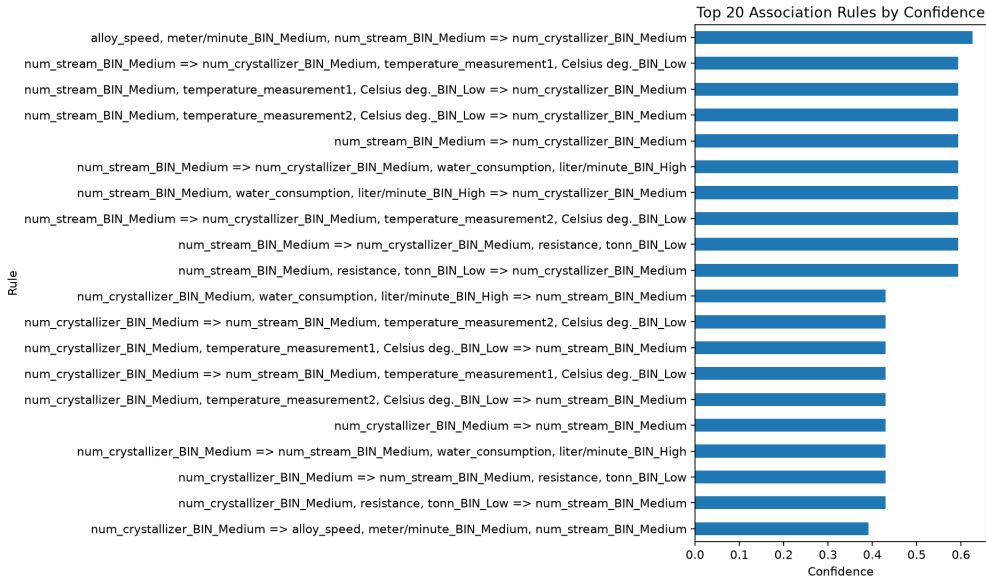


Fig. 22: Top 20 association rules by confidence

The top 20 rules ranked by confidence show high reliability, with values between 0.40 and 0.65. This means that when certain heavy material weights and high tonnage levels occur, the system is in a healthy **RUL** class in 40% to 65% of cases.

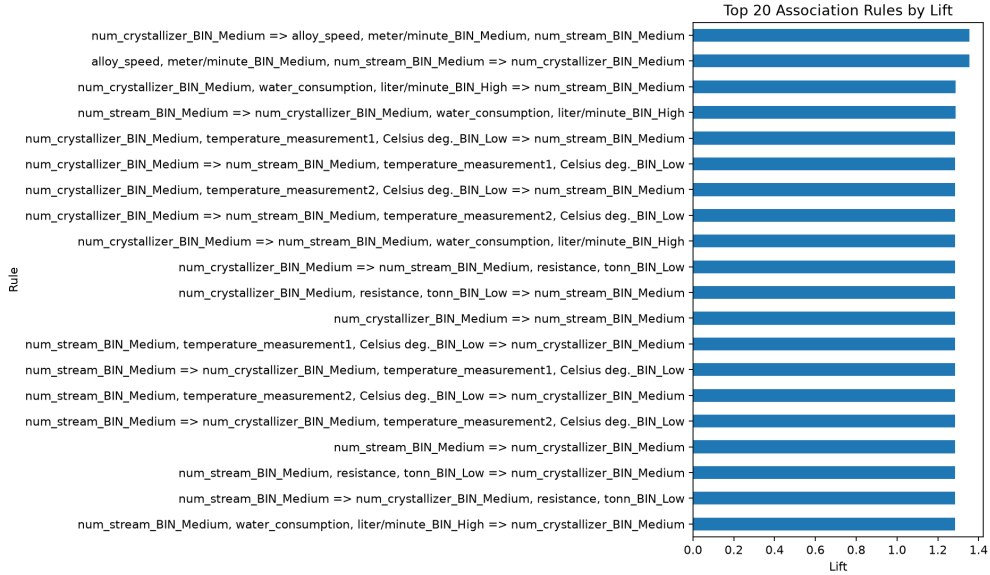


Fig. 23: Top 20 association rules by lift

The top 20 rules ranked by lift all have values above "1.2", reaching up to about "1.4". Since a lift greater than 1 indicates a real association beyond chance, these results show that these operating conditions are linked to machine behaviour.

As a conclusion for rule mining, **the results suggest that specific combinations of load and tonnage are good indicators of stable machine operation.**

5. Knowledge Discovery

5.1. Knowledge Discovery

First of all, let's define every field approached within this document. **Knowledge Discovery (KD)** is the process of extracting general, meaningful patterns from data rather than analysing isolated records [7]. Its objective is to transform raw data into structured and explainable knowledge that captures what is consistently true in the dataset.

KD focuses on identifying:

- **Patterns:** regular structures in data;
- **General rules:** valid beyond individual cases);
- **Stable relationships:** robust dependencies between variables.

5.2. Formal Concept Analysis

Formal Concept Analysis (**FCA**) is a mathematical framework for discovering structured knowledge in data [8]. It represents information as a set of **formal concepts**, each composed of:

- **Extent:** a set of objects;
- **Intent:** a set of shared attributes.

These concepts are organized into a **concept lattice**, which orders knowledge from general to specific concepts. **FCA** is useful in **KD** because it produces:

- Interpretable concept hierarchies;
- Logical implication rules (*if A then B*);
- Reduced and non-redundant knowledge representations.

Overall, **FCA** transforms raw data into structured conceptual systems, enabling the extraction of exact dependencies between attributes and supporting explainable knowledge discovery.

5.3. ToscanaJ and Elba

ToscanaJ is a conceptual data visualization tool used to display formal conceptual structures, such as concept lattices, derived from relational databases or in-memory data [2]. It provides a graphical interface for exploring conceptual schemas and supports both simple and nested lattice diagrams. Key capabilities include:

- Visualization of concept lattices using line diagrams;
- Interactive node selection and filtering of object sets;
- Visual encoding of data;
- Export of diagrams in formats such as SVG, PNG and JPEG;
- Integration with database views and external web-based attribute viewers.

Elba is a conceptual system editor used to construct formal contexts and conceptual schemas directly on top of relational databases. It supports database-aware modeling, allowing users to define scales, attributes and conceptual structures while interacting with underlying data sources. Key features include:

- Database connection support (JDBC, ODBC, MS Access);
- Scale generation tools for constructing conceptual attributes,
- Diagram editing and layout management based on structured grid systems;
- XML-based schema editing and export capabilities;
- SQL export for database integration and migration;
- Support for generating formal contexts and conceptual diagrams.

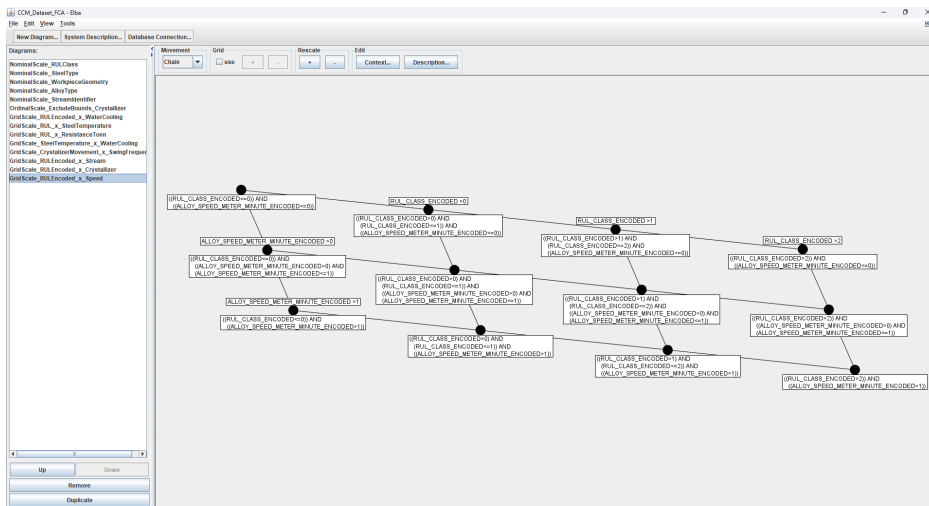


Fig. 24: Elba Application Sample

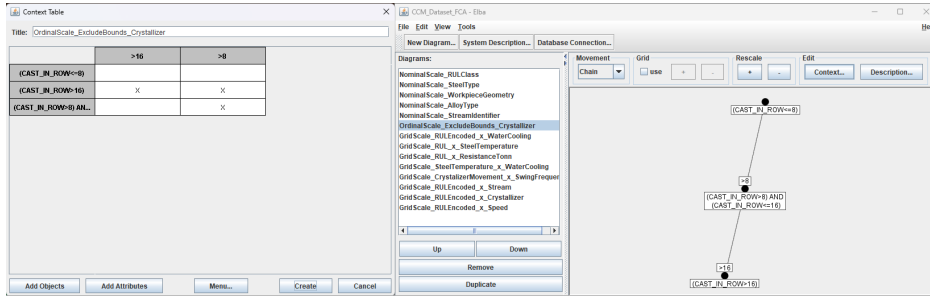


Fig. 25: Elba Application Sample

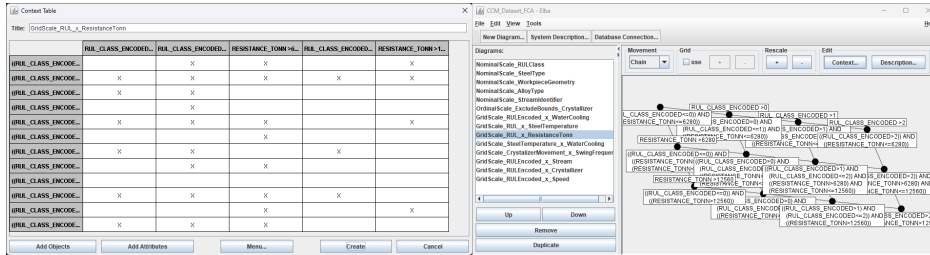


Fig. 26: Elba Application Sample

5.4. Conceptual Scales for FCA Analysis

The selected database for this analysis part is the preprocessed one. To enable **FCA** over the industrial dataset, it is necessary to define a set of **nominal, ordinal and grid based conceptual scales**. These scales discretize process variables into interpretable semantic categories (0 as Low, 1 as Medium and 2 as High for encoded variables; 0 as Critical, 1 as Low, 2 as Medium and 3 as Healthy for **RUL** class encoding), enabling the extraction of meaningful implication rules and concept lattices.

Scale Name	Type	Values / Structure
NominalScale_RUL_Class	Nominal	0 as Critical, 1 as Low, 2 as medium, 3 as Healthy
NominalScale_SteelType	Nominal	All processed steel types
NominalScale_WorkpieceGeometry	Nominal	150x150 or 180x180
NominalScale_AlloyType	Nominal	Open or closed
NominalScale_StreamIdentifier	Nominal	Identifier from 1 to 6
OrdinalScale.ExcludedBounds.CastInRow	Ordinal	$i=8 / i_8$ and $i=16 / i_{16}$ (or 1-8, 9-16, 17-24)
GridScale_RULEncoded_x_WaterCooling	Grid	RUL Class Encoded $\times$ Water Cooling Efficiency
GridScale_RUL_x_SteelTemperature	Grid	RUL Class Encoded $\times$ Steel Temperature Thermal Stress
GridScale_RULEncoded_x_ResistanceTonn	Grid	RUL Class Encoded $\times$ Resistance
GridScale_SteelTemperature_x_WaterCooling	Grid	Steel Temperature Thermal Stress $\times$ Water Cooling Efficiency
GridScale_RULEncoded_x_Stream	Grid	RUL Class Encoded $\times$ Stream Identifier
GridScale_RULEncoded_x_Crystallizer	Grid	RUL Class Encoded $\times$ Crystallizer Identifier (1-8, 9-16, 17-24)
GridScale_RULEncoded_x_Speed	Grid	RUL Class Encoded $\times$ Alloy Speed Meter per Minute

Table 6: Formal Concept Analysis scales constructed in Elba

The previous **nominal scales** are a basic chart used in Formal Concept Analysis (FCA). They separate data into flat unconnected categories and act like a simple summary that shows the percentage of items in each group. They do not provide particular hidden gems for knowledge discovery.

Here is what the four previous provided nominal scales expose:

- **Alloy Type Distribution:** This scale shows that the casting process almost exclusively uses the “open” alloy type (99.02%). The “close” type is very rare (0.98%);
- **RUL Class Distribution:** This scale shows a perfectly balanced dataset. The four categories (Critical, Low, Medium and Healthy) are almost equal in size, each holding roughly 24% to 26% of the data;
- **Steel Type Distribution:** This scale reveals a heavy preference for one specific material. The “Arm500” steel type makes up 79.86% of the records. The remaining data is spread across the other minor steel types;
- **Workpiece Geometry Distribution:** This scale tracks the physical shape of the cast steel. The larger “180×180” size is the most common at 87.4%, while the smaller “150×150” size makes up 12.6%.

More interesting results were obtained from the **grid scales**, which combine multiple attributes into a single analysis. These scales allow the exploration of interactions between process conditions and **RUL** classes, helping to identify patterns and relationships that are not visible when each attribute is analysed independently. As a result, the **grid scales provided a richer source of knowledge discovery** for the continuous casting process.

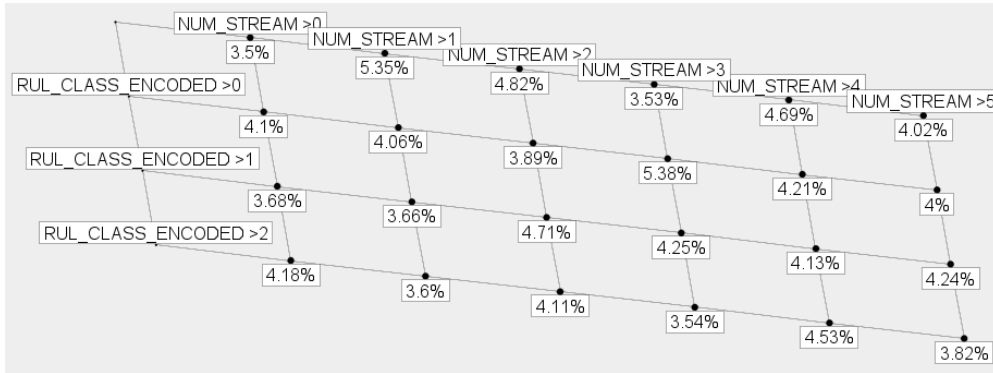
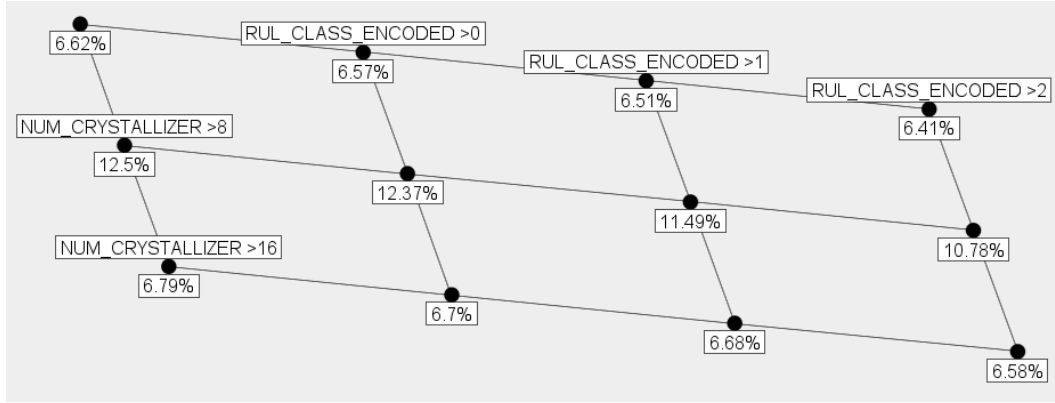


Fig. 28: ToscanaJ FCA Grid Scale Representation, **RUL** × **Stream**

In this grid scale, the columns represent the ordinal scaling of the **NUM_STREAM** attribute, while the rows represent the ordinal scaling of the **RUL_CLASS_ENCODED** attribute (row 1 represents value 0 meaning Critical, row 2 represents “> 0 and ≤ 1” as Low, “> 1 and ≤ 2” as Medium and “> 2” as Healthy). The intersection at each cell contains the percentage of objects distribution belonging to that specific combined concept node. Values are almost equally spread trough nodes, varying between 3-5%.

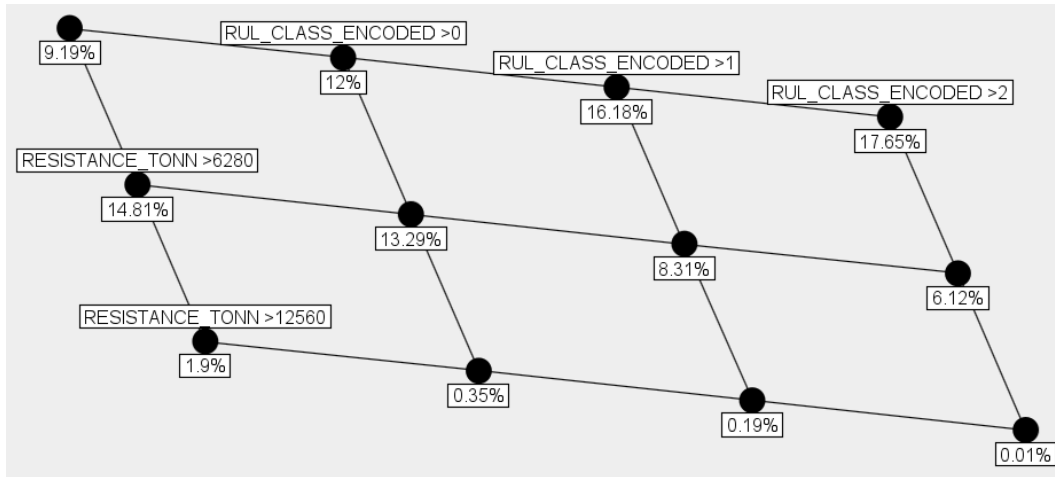
RUL / NUM.STREAM	> 0 & ≤ 1	> 1 & ≤ 2	> 2 & ≤ 3	> 3 & ≤ 4	> 4 & ≤ 5	> 5
RUL ≤ 0	3.50%	5.35%	4.82%	3.53%	4.69%	4.02%
RUL > 0 & ≤ 1	4.10%	4.06%	3.89%	5.38%	4.21%	4.00%
RUL > 1 & ≤ 2	3.68%	3.66%	4.71%	4.25%	4.13%	4.24%
RUL > 2	4.18%	3.60%	4.11%	3.54%	4.53%	3.82%

Table 7: Grid scale interaction between **NUM.STREAM** and **RUL.Class** distribution

Fig. 29: ToscanaJ FCA Grid Scale Representation, $RUL \times$ Crystallizer

The grid scale analysis shows a relationship between the crystallizer used in the casting process and the RUL of the sleeve. The crystallizers are divided into three identifier groups for an easier visualization (1–8, 9–16 and 17+). The first and third groups have a balanced distribution of RUL classes, indicating a normal and stable degradation process.

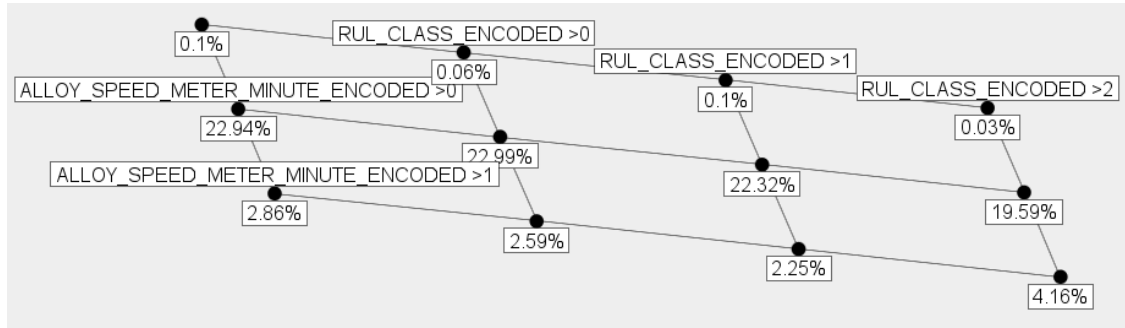
The second group of crystallizers (9–16) shows a different pattern. This group processes a larger share of the production data and contains a higher proportion of Critical RUL cases compared to Healthy cases (column 3). This result suggests that crystallizers 9–16 may experience higher stress or less favourable operating conditions, leading to faster sleeve degradation. Therefore, this group represents a potential bottleneck in the production process and should be investigated to improve equipment lifetime and reduce maintenance costs.

Fig. 30: ToscanaJ FCA Grid Scale Representation, $RUL \times$ Stream

The previous grid scale analyses the relationship between the amount of processed steel tonnage and the RUL of the sleeve. Sleeves with low processed tonnage are mostly found in the Healthy RUL class, indicating that the equipment performs well during the early stages of its life cycle.

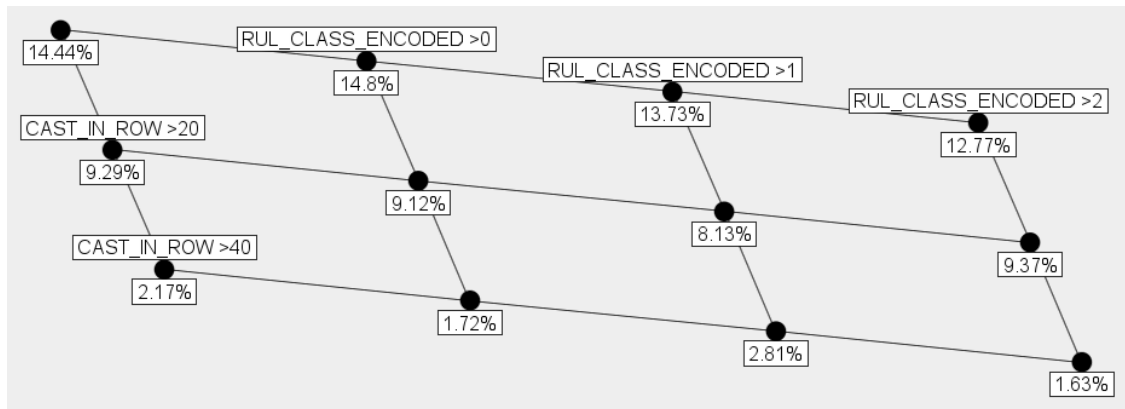
As the processed tonnage increases, the distribution shifts toward the Critical and Low RUL classes. This indicates that sleeve degradation becomes more pronounced after a certain usage level. The middle tonnage range contains the largest proportion of Critical sleeves, showing that wear increases significantly during the mid life stage of the equipment.

The analysis also shows that **very few sleeves remain in operation after processing more than 12,560 tons of steel**. Among these cases, almost none belong to the Healthy class. This suggests that most sleeves reach a critical condition and are replaced.

Fig. 31: ToscanaJ FCA Grid Scale Representation, $RUL \times$ Crystallizer

The previous grid scale shows the relationship between casting speed and RUL . Most production data belongs to the medium speed category, indicating that this is the normal operating speed of the casting process. In this range, the distribution contains slightly more Critical sleeves than Healthy sleeves, suggesting that equipment wear increases during regular operation.

The high-speed category shows a larger proportion of Healthy sleeves and a smaller proportion of Critical sleeves, suggesting that high casting speeds are probably used when the sleeve is in good condition. As the sleeve degrades and moves toward lower RUL , the casting speed is reduced to maintain process stability and product quality.

Fig. 32: ToscanaJ FCA Grid Scale Representation, $RUL \times$ Cast in row

The grid scale analysis shows the relationship between sleeve health RUL class and the length of continuous casting sequences (CAST_IN_ROW). As observed, most production happens in short sequences (0–20 casts), while fewer cases occur in medium (20+) and long (40+) sequences. In short sequences, the data shows more Critical sleeves than Healthy ones, **indicating higher wear during frequent stops and restarts.**

When the casting process runs for longer continuous sequences, the wear becomes more balanced between RUL classes, with similar proportions of Critical and Healthy sleeves. This suggests that **stable long production runs reduce degradation.**

6. Attribute Exploration

6.1. Attribute Exploration

First of all, let's define what **Attribute Exploration** is. **Attribute Exploration (AE)** is a method from Formal Concept Analysis (**FCA**) used to discover all valid attribute dependencies in a dataset. It starts from a formal context, which is a table describing which objects have which attributes. The goal is to identify implication rules of the form $(A \text{ to } B)$, meaning that whenever an object has all attributes in (A) , it must also have all attributes in (B) [17].

The method works by systematically generating candidate implications and checking whether they are valid in the dataset. If a candidate implication is correct, it is added to a growing set of rules, else if it is not correct, a counter example is required. This counterexample is then added to the dataset to refine the analysis.

A key idea in **AE** is that the process continues until no new valid implications can be found. The resulting set of implications forms a complete and non redundant description of the relationships between attributes. The method is strongly connected to closure systems and logical reasoning.

6.2. Attribute Exploration **FCA** Setup

The selected attributes are represented as a formal context $\mathbb{K} = (G, M, I)$, where G is the set of process observations, M is the set of attributes and I is the binary relation indicating attribute presence.

Attribute	Meaning
A1	Alloy type category (Open, Close)
A2	Cast in row length (Low, Medium, High)
A3	Alloy speed level (Low, Medium, High)
A4	Steel weight category (Low, Medium, High)
A5	Crystallizer index group (Low, Medium, High)
A6	RUL class (Critical, Low, Medium, Healthy)

Table 8: Semantic description of selected attributes

In the following **AE** table, each value 1 indicates that the object contains the corresponding attribute, while 0 indicates absence.

Object	A1	A2	A3	A4	A5	A6
Sys1	1	0	1	1	0	1
Sys2	1	1	0	1	1	0
Sys3	0	1	1	0	1	0
Sys4	1	0	0	0	1	0
Sys5	0	1	1	1	0	1

Table 9: Binary formal context (A1–A6 attributes)

Attribute exploration uses the closure operator X'' to determine stable attribute sets.

Attribute Set	Closure X''	Result
{A3}	{A6}	RUL dependency appears
{A2}	{A6}	Stability effect observed
{A5}	{A6}	Equipment group effect
{A3, A2}	{A6}	Strong combined dependency

Table 10: Example closure results from attribute exploration

The attribute exploration process produces a set of implications that approximate the stem base.

Implication	Interpretation
A3 (High speed) $\rightarrow$ A6 (Healthy/Medium)	High speed is used under stable operating conditions
A2 (Long cast-in-row) $\rightarrow$ A6 (Healthy/Medium)	Long stable sequences indicate low degradation risk
A2 (Short cast-in-row) $\rightarrow$ A6 (Critical/Low)	Frequent stops increase thermal stress and wear
A5 (High crystallizer index) $\rightarrow$ A6 (Medium/Low)	Certain equipment groups show higher stress levels
A3 (Low speed) $\wedge$ A2 (Short row) $\rightarrow$ A6 (Critical)	Unstable operating regime strongly linked to failure states

Table 11: Extracted implication set (stem base approximation)

The implication system can be summarized as:

- Single-factor rules:
 - $A2 \rightarrow A6$
 - $A3 \rightarrow A6$
 - $A5 \rightarrow A6$
- Combined interaction rule:
 - $A2 \wedge A3 \rightarrow A6$
- Directional degradation pattern:
 - Short operation cycles map to Critical **RUL** states

The implication system shows that A6 (**RUL** class) is the main target of the closure process. Most attribute sets are extended toward A6, which means A6 is the main result attribute in the **FCA** model. There are no strong direct rules between process variables, but instead, most rules lead toward the equipment state.

This means the formal context mainly works as a classification system, where process attributes are used to predict a single outcome attribute.

7. Triadic Knowledge Discovery

7.1. Triadic Formal Concept Analysis

In previous section 5 normal FCA was performed on two dimensions. Triadic Formal Concept Analysis (**Triadic Formal Concept Analysis (3FCA)**) is an extension of standard FCA that works with data structured in three dimensions instead of two. While classical FCA connects **objects and attributes**, 3FCA adds a third component called **conditions** [18].

A triadic context is defined by three sets together with a relation that links all of them: **objects, attributes and conditions**. A relation entry means that an object has a specific attribute under a specific condition. This structure can be represented as a three dimensional data cube or as a collection of two dimensional tables, each corresponding to one condition.

In 3FCA the main analytical unit or relation between sets is the **triconcept**. Every object in the set has all attributes under all conditions in the group. Triconcepts describe stable patterns that hold across all three dimensions at the same time. This allows more detailed pattern discovery compared to classical FCA, especially in domains where behaviour changes across different environments or situations.

7.2. 3FCA on Continuous Casting Data

3FCA is applied to the continuous casting dataset to model relationships between process variables, equipment configuration and operating regimes. The dataset is represented as a triadic context $\mathbb{K} = (K_1, K_2, K_3, Y)$, expressed as follow:

- K_1 (Objects): casting production records;
- K_2 (Attributes): process and equipment variables;
- K_3 (Conditions): discretized operational regimes derived from the FCA grid analysis;
- Y (Relation): contains triples (g, m, c) meaning that a production record g has attribute m under condition c .

The triadic structure is constructed using previously validated FCA findings in section 5:

- Attributes (K_2):
 - Alloy speed (Low, Medium, High);
 - Cast-in-row length (Low, Medium, High);
 - Num crystallizer group (Low, Medium, High as 1-8, 9-16, 17+);
 - Steel weight (Low, Medium, High);
 - RUL class (Critical, Low, Medium, Healthy).
- Conditions (K_3) derived from FCA grid structure:
 - Low stability regime (short cast-in-row, low speed);
 - Medium stability regime (mixed operational conditions);
 - High stability regime (long cast-in-row, high speed).

Conditions represent combined operational states rather than single variables.

The FCA grid analysis provides the following validated structure:

- Low stability regimes are strongly associated with Critical and Low RUL classes;
- Medium regimes show mixed distributions across RUL classes;
- High stability regimes are strongly associated with higher RUL classes (Healthy / Medium).

In addition, as studied in section 5, the crystallizer group (9-16) shows a higher proportion of Critical cases, indicating a localized stress region in the production system.

The triadic relation Y is illustrated using representative validated patterns:

Attribute	Condition	RUL State	Relation
High Alloy Speed	High Stability	Healthy	Valid
Long Cast-in-Row	High Stability	Medium / Healthy	Valid
Short Cast-in-Row	Low Stability	Critical	Valid
Low Alloy Speed	Low Stability	Critical	Valid
Crystallizer 9-16	Medium Stability	Critical (higher share)	Valid

Table 12: Triadic incidence derived from validated FCA patterns

A **triconcept** is a maximal triple (A_1, A_2, A_3) such that all casting records in A_1 share attributes A_2 under all conditions in A_3 . In this dataset, triconcepts correspond to stable operational regimes:

- High stability runs, High speed + Long cast-in-row, Healthy RUL;
- Low stability runs, Low speed + Short cast-in-row, Critical RUL;
- Medium stability runs, Mixed parameters, Mixed RUL states;
- Crystallizer 9-16 runs, Stress-prone configuration, Higher Critical ratio.

The triadic structure confirms that equipment degradation is driven by combined operational regimes rather than isolated variables. **RUL class behaves as a dependent outcome of stability conditions**, high stability regimes correspond mainly to Healthy or Medium RUL, while low stability regimes correspond to Critical RUL. Furthermore, the crystallizer grouping introduces an additional structural effect where the 9-16 group increases the probability of Critical states.

8. Temporal Knowledge Discovery

8.1. Temporal Concept Analysis

Temporal Concept Analysis (TCA) is an extension of FCA that is used to **analyse how structured knowledge changes over time**. While FCA describes data in a static form using objects, attributes and formal concepts, **TCA adds a temporal dimension** [19]. This allows data to be observed at different time points, called time granules, where each time point is described by a formal context representing the state of the system at that moment.

In TCA, each time granule is mapped to a formal concept in a concept lattice, which represents the state of the system at that time. By connecting consecutive time points, transitions between concepts are formed, creating a state-transition structure. This structure shows how the system evolves from one conceptual state to another over time, making it possible to analyse changes in a systematic way.

A key idea in TCA is the life-track, which describes the evolution of an individual object across time by mapping each time point to a corresponding formal concept. This produces a trajectory through the concept lattice that can be used to study patterns such as stability, change or repetition.

8.2. Temporal Concept Analysis in Continuous Casting Data

A **Conceptual Time System (CTS)** is defined as $T = (T_t, T_e)$, where T_t represents the temporal structure and T_e represents the event context. For the continuous casting dataset as $T_t = (G_T, <)$, where:

- G_T is the set of timestamps extracted from *datetime_combined*;
- $<$ is the chronological successor relation between timestamps.

The event context is defined as $K_e = (G_T, M_e, I_e)$, where G_T are the temporal observations, M_e are the FCA attributes and I_e is the incidence relation.

The attributes used in the temporal analysis are derived from the **FCA** scales: **RUL** class, Processed steel tonnage, Alloy speed, Cast in row and Crystallizer group. To illustrate the temporal structure used in **TCA**, a sleeve trajectory is represented as a sequence of temporal granules. Each granule corresponds to a timestamped observation and is associated with a set of discretised attributes derived from **FCA** scaling of process variables at that time. These attributes represent a discretised **FCA** scaling of continuous process variables, yielding a binary incidence representation suitable for temporal concept construction.

Table 13: Example event context for a sleeve trajectory

Time Granule	Healthy	Low Tonnage	Medium Tonnage	High Speed	Critical
g_1	×	×		×	
g_2	×	×		×	
g_3			×	×	
g_4			×		
g_5			×		
g_6					×

The temporal scale in Table 14 represents the ordering relation $<$ over the set of time granules G_T , while Table 13 defines the event context $K_e = (G_T, M_e, I_e)$ describing the attribute incidence relation at each temporal observation. Together, they constitute the foundational structure of the Conceptual Time System used in **TCA**.

Table 14: Temporal scale describing the chronological order

$<$	g_1	g_2	g_3	g_4	g_5	g_6	g_7
$\leq g_1$	×						
$\leq g_2$	×	×					
$\leq g_3$	×	×	×				
$\leq g_4$	×	×	×	×			
$\leq g_5$	×	×	×	×	×		
$\leq g_6$	×	×	×	×	×	×	
$\leq g_7$	×	×	×	×	×	×	×

Table 13 presents a simplified temporal context derived from a single sleeve trajectory. The rows correspond to temporal observations, while the columns correspond to **FCA** derived attributes defining the observed process conditions. This representation should not be interpreted as a complete dataset reconstruction. Instead, it serves as a formal example of the event context used within **TCA** for constructing temporal concepts and deriving life-tracks.

For every time granule $g \in G_T$, a conceptual state $\sigma(g) \in \Sigma$ is assigned based on the observed **FCA** attribute configuration at time g .

State	Concept Description
S_1	Healthy sleeve, low accumulated tonnage, high casting speed
S_2	Medium sleeve condition, increasing tonnage, normal casting speed
S_3	Low sleeve condition, reduced casting speed, increasing wear
S_4	Critical sleeve, high accumulated tonnage, maintenance required

Table 15: Representative conceptual states obtained from **FCA** knowledge

These states are not manually defined process phases. They are abstractions derived from the [FCA](#) concepts and their attribute combinations. The set of all states forms $\Sigma = \{\sigma(g) \mid g \in G_T\}$, which represents the conceptual state space of the continuous casting process. Formally, each state corresponds to a formal concept induced by a time granule $\sigma(g) = (\{g\}'', \{g\}') \in \mathcal{B}(K_e)$.

Given consecutive time granules $g_i < g_{i+1}$, a transition is defined as: $\sigma(g_i) \rightarrow \sigma(g_{i+1})$. The collection of all transitions defines the state-transition relation $\rightarrow \subseteq \Sigma \times \Sigma$, yielding the state-transition graph $S = (\Sigma, \rightarrow)$. The temporal ordering is independent from the concept lattice order. A transition is produced by time succession and not by concept specialization.

The dataset contains repeated observations of the same sleeve. Therefore, sleeves can be considered temporal objects.

The **Conceptual Time System with actual Objects and a Time relation (CTSOT)** structure is (P, G_T, ρ, K_e) , where:

- P is the set of sleeves;
- G_T is the set of timestamps;
- ρ associates sleeves with observation times;
- K_e is the [FCA](#) event context.

Each sleeve generates a temporal trajectory through the conceptual state space. For a sleeve p , the life-track is defined as $life_p : G_p \rightarrow \mathcal{B}(K_e)$, where G_p contains all timestamps associated with the sleeve. Using the [FCA](#) results, the typical life-track observed in the dataset is $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4$. This trajectory represents the progressive degradation of the sleeve.

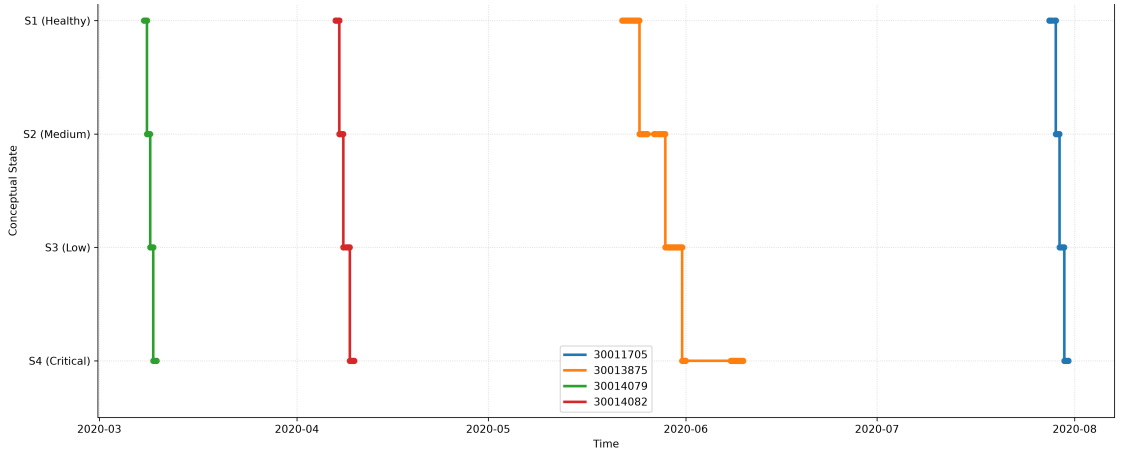


Fig. 33: [TCA](#) empirical life tracks $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4$

Initially, sleeves operate with low accumulated tonnage and higher casting speeds. As production continues, processed tonnage increases and sleeve condition deteriorates. The process eventually reaches a critical state where maintenance or replacement becomes necessary.

The [FCA](#) analysis showed that healthy sleeves are mainly associated with low processed tonnage. Temporal analysis suggests that these states appear at the beginning of sleeve life-tracks. The [FCA](#) analysis also showed that critical sleeves become more frequent as accumulated tonnage increases. In [TCA](#), this corresponds to transitions toward degradation states later in the trajectory.

Casting speed provides another temporal indicator. Healthy sleeves are more frequently associated with high casting speeds, while degraded sleeves are associated with lower speeds. This suggests a gradual transition from high speed operation to more conservative operating conditions as wear progresses.

The **FCA** analysis also revealed that short casting sequences contain more critical sleeves than healthy sleeves. From a temporal perspective, frequent production interruptions may accelerate transitions toward degraded states. Longer continuous sequences appear to maintain more stable trajectories and slower degradation.

Finally, crystallizers belonging to the 9-16 group contain a larger proportion of critical sleeves. In the temporal state space, trajectories involving these crystallizers are expected to reach degradation states more frequently than trajectories associated with other crystallizer groups.

Several temporal patterns emerge from the analysis:

- Healthy states usually occur before Critical states;
- Increasing processed tonnage precedes degradation states;
- Reduced casting speed frequently appears before Critical states;
- Long continuous casting runs are associated with more stable trajectories;
- Crystallizers 9-16 appear more frequently in degradation trajectories.

These patterns are consistent with the **FCA** knowledge previously extracted from the dataset.

The **FCA** analysis identified the conceptual relationships between sleeve health and process variables. **TCA** extends these results by introducing chronological ordering between observations.

9. Conclusion

9.1. Research Paper Summary

This research analysed data from a Continuous Casting Machine (**CCM**). A **CCM** is an industrial system used to convert liquid steel into solid billets. The main focus of the study was the mold **sleeve**, a water-cooled copper component that initiates steel solidification. Mold sleeves degrade over time due to thermal load and mechanical friction.

The primary objective was to analyse and estimate the Remaining Useful Life (**RUL**) of mold sleeves. **RUL** represents the amount of steel (in tons) that can be processed before a sleeve must be replaced. To achieve this objective, several knowledge discovery and data mining methods were applied to extract patterns and rules from real production data.

9.2. Knowledge Discovery Overview

Formal Concept Analysis (**FCA**) is a mathematical method used to group objects based on shared attributes. In this study, **FCA** was used to transform raw sensor data into structured concept lattices. Key advantages include:

- **Rule extraction:** **FCA** generates clear logical relationships between process variables and outcomes;
- **Interpretability:** The resulting structures are human readable and can be directly interpreted by engineers;
- **Data reduction:** Irrelevant or redundant information is filtered, producing compact representations of the data.

Triadic Formal Concept Analysis (**3FCA**) extends **FCA** by introducing a third dimension representing conditions or contexts. Key features include:

- **Context dependent analysis:** Machine behavior can be evaluated across different operating regimes;
- **Multi factor patterns:** Complex interactions between multiple variables can be identified.

Temporal Concept Analysis (TCA) extends FCA by incorporating time. It enables the analysis of how system states evolve over time. Key results:

- **State tracking:** Transitions from healthy to degraded states can be observed;
- **Lifecycle modelling:** Degradation trajectories of equipment can be reconstructed over time.

The combined application of these methods produced several key findings in the continuous casting process:

- **Speed regulation:** Higher casting speeds are generally used when mold sleeves are new. As wear increases, operating speed is reduced to maintain process stability and product quality;
- **Operational bottlenecks:** Crystallizers numbered 9 to 16 show higher stress levels and faster degradation compared to other units. These components represent critical maintenance points;
- **Stability effects:** Long continuous production sequences reduce wear rates, while frequent stops and restarts increase thermal stress and accelerate degradation.

9.3. Final Statement

The application of structured knowledge discovery methods provides a clear link between raw industrial sensor data and actionable maintenance insights. These methods support improved understanding of equipment degradation and enable more effective decision making in steel production systems.

9.4. Turnitin Similarity Check

Turnitin is a widely used academic integrity platform used to evaluate the originality of written documents. It compares submitted texts against a large database of academic publications, web sources and previously submitted student papers. The system generates a similarity report that highlights overlapping content and provides a similarity score.

A similarity score of approximately 10% is considered very good in most academic contexts. This level indicates that only a small portion of the text matches existing sources and these matches are typically due to standard terminology, references or commonly used technical expressions. Such a result generally reflects a high level of originality in the written work while still maintaining appropriate academic referencing standards.

More information about the platform is available at: [TurnItIn](https://turnitin.com)

The screenshot displays the Turnitin Assignment Dashboard on the left and the Match Overview report on the right.

Turnitin Assignment Dashboard (Left):

- Header: Turnitin - Assignment Dashboard
- Navigation: Class Portfolio, My Grades, Discussion, Calendar
- Section: NOW VIEWING: HOME > ICA1 > REFERATE
- About this page: This is your assignment dashboard. You can upload submissions for your assignment from here. When a submission has been processed you will be able to download a digital receipt, view any grades and similarity reports that have been made available by your instructor.
- Table:

Paper Title	Uploaded	Grade	Similarity
KD_Report_Paper_Todoran_New_TURNITIN_TODO	06/21/2026 4:42 PM	--	10%

Match Overview Report (Right):

- Header: Match Overview
- Overall Similarity Score: 10%
- Table of Matches:

Rank	Source	Similarity
1	"Conceptual Knowledge..." Publication	<1%
2	Dothang Truong, "Dem..." Publication	<1%
3	K. V. Sambasivaran, An... Publication	<1%
4	ijcsmc.com	<1%
5	Submitted to Valencia...	<1%
6	Submitted to University...	<1%
7	Submitted to Taylor's E...	<1%
8	download.bibis.ir	<1%
9	theses.hal.science	<1%
10	Submitted to Babes-B...	<1%
11	Jan Gunis, Lubomir Šn...	<1%
12	ceur-ws.org	<1%
13	synopsis.koreamed.org	<1%

Fig. 34: TurnItIn Similarity Score Overview

References

- [1] Agrawal, R.S., Srikant, R., 1994. R. fast algorithms for mining association rules, in: Proceedings of the 20th international conference on very large data bases, vldb, pp. 487–499.
- [2] Becker, P., Correia, J.H., 2005. The toscanaj suite for implementing conceptual information systems, in: Formal concept analysis: Foundations and applications. Springer, pp. 324–348.
- [3] Bishop, C.M., 2006. Pattern Recognition and Machine Learning. Springer.
- [4] Breiman, L., 2001. Random forests. Machine learning 45, 5–32.
- [5] Chai, T., Draxler, R.R., 2014. Root mean square error (rmse) or mean absolute error (mae)?—arguments against avoiding rmse in the literature. Geoscientific model development 7, 1247–1250.
- [6] Ester, M., Kriegel, H.P., Sander, J., Xu, X., et al., 1996. A density-based algorithm for discovering clusters in large spatial databases with noise, in: kdd, pp. 226–231.
- [7] Fayyad, U., Piatetsky-Shapiro, G., Smyth, P., 1996. From data mining to knowledge discovery in databases. AI Magazine 17, 37–54.
- [8] Ganter, B., Wille, R., 1999. Formal Concept Analysis: Mathematical Foundations. Springer.
- [9] Guthrie, R.I., Isac, M.M., 2022. Continuous casting practices for steel: Past, present and future. Metals 12, 862.
- [10] Han, J., Kamber, M., Pei, J., 2022. Data Mining: Concepts and Techniques. 4th ed., Morgan Kaufmann.
- [11] Hyndman, R.J., Koehler, A.B., 2006. Another look at measures of forecast accuracy. International journal of forecasting 22, 679–688.
- [12] Jolliffe, I.T., Cadima, J., 2016. Principal component analysis: a review and recent developments. Philosophical transactions. Series A, Mathematical, physical, and engineering sciences 374, 20150202.
- [13] Katser, I., Mezhev, M., Mozolin, I., Travnikov, V., Kozitsin, V., 2023. Rul dataset from continuous casting machine. URL: <https://www.kaggle.com/ds/4014009>, doi:10.34740/KAGGLE/DS/4014009.
- [14] Liu, F.T., Ting, K.M., Zhou, Z.H., 2008. Isolation forest, in: 2008 eighth IEEE international conference on data mining, IEEE. pp. 413–422.
- [15] Powers, D.M., 2020. Evaluation: from precision, recall and f-measure to roc, informedness, markedness and correlation. arXiv preprint arXiv:2010.16061 .
- [16] Russell, S., Norvig, P., 2021. Artificial Intelligence: A Modern Approach. 4th ed., Pearson.
- [17] Stumme, G., 1996. Attribute exploration with background implications and exceptions, in: Data Analysis and Information Systems: Statistical and Conceptual Approaches Proceedings of the 19th Annual Conference of the Gesellschaft für Klassifikation eV University of Basel, March 8–10, 1995, Springer. pp. 457–469.
- [18] Wille, R., 1995. The basic theorem of triadic concept analysis. Order 12, 149–158.
- [19] Wolff, K.E., 2001. Process representation in temporal concept analysis. Conceptual Structures: Extracting and Representing Semantics. Contributions to ICCS , 211–213.
- [20] Wolff, K.E., 2005. States, transitions and life tracks in temporal concept analysis, in: Formal Concept Analysis, Springer. pp. 127–148.

Acronyms

3FCA Triadic Formal Concept Analysis. [1](#), [27](#), [31](#)

AE Attribute Exploration. [25](#)

AI Artificial Intelligence. [2](#)

CCM Continuous Casting Machine. [1–3](#), [7](#), [31](#)

CTS Conceptual Time System. [28](#)

DA Data Analysis. [7](#)

DBSCAN Density-Based Spatial Clustering of Applications with Noise. [1](#), [14](#), [15](#)

DM Data Mining. [2](#)

FCA Formal Concept Analysis. [1–3](#), [6](#), [7](#), [17–20](#), [22](#), [25–32](#)

IF Isolation Forest. [1](#), [16](#), [17](#)

KD Knowledge Discovery. [7](#), [18](#), [19](#)

KDD Knowledge Discovery in Databases. [2](#)

ML Machine Learning. [2](#), [3](#), [14](#)

PCA Principal Component Analysis. [1](#), [14–16](#)

RFC Random Forest Classifier. [1](#), [12](#)

RFR Random Forest Regressor. [1](#), [11–13](#)

RUL Remaining Useful Life. [1](#), [3–12](#), [14](#), [18](#), [20](#), [22–29](#), [31](#)

SE Software Engineer. [2](#)

SI Steel Industry. [2](#)

SL Supervised Learning. [11](#), [14](#)

TCA Temporal Concept Analysis. [1–3](#), [28–32](#)

UL Unsupervised Learning. [14](#)